TECHNOLOGY

Engineers and their triumphs

F. M. Holmes

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Transcriber’s Notes

Hyphenation has been standardised.

The Transcriber has constructed a ‘List of Illustrations’, as none was supplied.

Page 132 — changed possibilites to =possibilities=

[Illustration: THE TOWER BRIDGE, LONDON, SHOWING THE BASCULES RAISED.]

ENGINEERS

AND

THEIR TRIUMPHS:

THE STORY OF THE LOCOMOTIVE—THE STEAMSHIP—BRIDGE BUILDING—TUNNEL MAKING.

BY

F. M. HOLMES,

AUTHOR OF “FOUR HEROES OF INDIA,” ETC.

[Illustration]

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[Illustration]

PREFACE.

Without attempting to be exhaustive, this little book aims at describing in a purely popular and non-technical manner some of the great achievements of engineers, more particularly during the nineteenth century.

The four departments chosen have been selected not in pursuance of any comprehensive plan, but because they present some of the more striking features of constructional effort. The term Engineering, however, includes the design and supervision of numerous works, such as roads and canals, docks and break-waters, machinery and mining, as well as steam-engines and steamships, bridges and tunnels.

Information, in certain cases, has been gained at first-hand, and I have to acknowledge the courtesy of the managers of the Cunard and White Star Steamship Companies, Messrs. Maudslay, Sons & Field, and others, in supplying various particulars.

The narrative concerning Henry Bell and the steamship _Comet_, and of his connection with Fulton, is chiefly based on a letter from Bell himself in the _Caledonian Mercury_ in 1816.

The statement that Mr. Macgregor Laird was so largely instrumental in founding the British and American Steam Navigation Company is made on the authority of his daughter, Miss Eleanor Bristow Laird. An article on “The Genesis of the Steamship,” which I wrote in the _Gentleman’s Magazine_, brought a letter from that lady in which she declares that her father was the prime mover in founding the Company. He had had experience, in the Niger Expedition of 1832-33, of the behaviour of steamships both at sea and in the river, and from the date of his return to England she asserts he advocated the establishment of steam communication between England and America, against the active opposition of Dr. Lardner and others. “Macgregor Laird’s claim to the foremost place amongst all those (not excepting Brunel) who worked for the same object,” writes Miss Laird, “was clearly shown in a letter from the late Mr. Archibald Hamilton of 17 St. Helen’s Place, E.C., to the editor of the _Shipping and Mercantile Gazette_, in which paper it was published on 15th May, 1873.”

It is not a little curious to note how, in many of these great undertakings, several minds seem to have been working to the same end at about the same time. It was so with George Stephenson and others with regard to the locomotive, with Miller and Symington, Bell and Fulton, with regard to the steamship, with Laird and Brunel as regards transatlantic steam navigation, with Robert Stephenson and William Fairbairn as regards the tubular bridge.

This volume does not seek to be the special advocate of any, or to enter into any minute details, but simply endeavours to gather up the more salient features and weave them into a connected and popular narrative.

F. M. HOLMES.

[Illustration]

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[Illustration]

ENGINEERS AND THEIR TRIUMPHS.

THE STORY OF THE LOCOMOTIVE.

CHAPTER I.

FIRST STEPS.

“I think I could make a better engine than that.”

“Do you? Well, some’ing’s wanted; hauling coal by horses is very expensive.”

“Ay, it is, and I think an engine could do it better.”

“Mr. Blackett’s second engine burst all to pieces; d’ye mind that?”

“How came that about?”

“Tommy Waters, who put it together, could not make it go, so he got a bit fractious and said she should move. He did some’ing to the safety-valve and she did begin to work, but then she burst all to pieces.”

“Ay, ay, but this one is an improvement.”

“It had need be. Even the third was a perfect plague.”

“What! you mean Mr. Blackett’s third engine?”

“Ay. It used to draw eight or nine truck loads at about a mile an hour, or a little less; but it often got cranky and stood still.”

“Stood still!”

“Ay; we thought she would never stick to the road, so we had a cogged wheel to work into a rack-work rail laid along the track, and somehow she was always getting off the rack-rail.”

“And now you find that the engine is heavy enough herself to grip the rail.”

“Ay, that was Will Hedley’s notion; he’s a viewer at the colliery. And it is a great improvement. Why, that third engine, I say, was a perfect nuisance. Chaps used to sing out to the driver: ‘How do you get on?’”

“‘Get on,’ sez he, ‘I don’t get on; I on’y get off!’”

“It was always goin’ wrong, and horses was always having to be got out to drag it along.”

“How did Hedley find out that a rack-rail was not needful?”

“Well, he had a framework put upon wheels and worked by windlasses which were geared to the wheels. Men were put to work these windlasses which set the wheels going; and, lo and behold, she moved! The wheels, though smooth, kept to the rails, though they were smooth also, and the framework went along without slipping. ‘Crikey!’ says Hedley, ‘no cogged wheels, no chains, no legs for me! We can do without ’em all. Smooth wheels will grip smooth rails.’ And he proved it too by several

experiments.”

“Then Mr. Blackett had this engine built?”

“Ay, and it be, as you say, a great improvement. But that steam blowing off there, after it have done its work, frights the horses on the Wylam Road ter’ble, and makes it a perfect nuisance.”

“Has nothing been done to alter it?”

“Mr. Blackett has given orders to stop the engine when any horses comes along, and the men don’t like that because it loses time. He thinks he is going to let the steam escape gradual like, by blowing it off into a cask first.”

“Umph! very wasteful.”

“Oh, ay; it be wasteful; and many a one about here sez of Mr. Blackett that a fool and his money are soon parted.”

“No,” said the first speaker, shaking his head thoughtfully, “Mr. Blackett is no fool. But I think I could build a better engine than that.”

[Illustration: GEORGE STEPHENSON.]

The tone in which these words were uttered was not boastful, but quiet and thoughtful.

“You are Geordie Stephenson, the engine-wright of the Killingworth Collieries, ’beant you?”

“Ay; and we have to haul coal some miles to the Tyne where it can be shipped. So you do away with all rack-work rails and all cogged wheels, do you?”

“Ay, ay, Geordie, that’s so—smooth wheels on smooth rails.”

This conversation, imaginary though to some extent it be, yet embodies some important facts. Jonathan Foster, Mr. Blackett’s engine-wright, informed Mr. Samuel Smiles, who mentions the circumstance in his “Lives of the Engineers,” that George Stephenson “declared his conviction that a much more effective engine might be made, that should work more steadily and draw the load more effectively.”

Geordie had studied the steam-engine most diligently. Born at Wylam—some eight miles distant from Newcastle, about thirty years previously—he had become a fireman of a steam-engine and had been wont to take it to pieces in his leisure. He was now thinking over the subject of building a locomotive engine, and he decided to see what had already been accomplished. He would profit by the failures and successes of others. So he went over to Wylam to see Mr. Blackett’s engines, and to Coxlodge Colliery to see Mr. Blenkinsop’s from Leeds; and here again it is said, that after watching the machine haul sixteen locomotive waggons at a speed of about three miles an hour, he expressed the opinion that “he thought he could make a better engine than that, to go upon legs.”

A man named Brunton did actually take out a patent in 1813 for doing this. The legs were to work alternately, like a living creature’s. The idea which seems to have

troubled the early inventors of the locomotive, was that smooth wheels would not grip smooth rails to haul along a load. And it was Blenkinsop of Leeds who took out a patent in 1811 for a rack-work rail into which a cog-wheel from his engine should work.

Thus William Hedley's idea of trusting to the weight of the engine to grip the rails, and abolishing all the toothed wheels and legs and rack-work for this purpose on a fairly level rail, was the first great step toward making the locomotive a practicable success.

[Illustration: "PUFFING BILLY," THE OLDEST LOCOMOTIVE ENGINE IN EXISTENCE.

(_At present in South Kensington Museum._)]

The idea that Stephenson invented the locomotive is a mistake. But just as James Watt improved the crude steam pumps and engines he found in existence, so George Stephenson of immortal memory developed and made practicable the locomotive. For, in spite of Hedley's discovery or invention, all locomotives were partial failures until Stephenson took the matter in hand.

Nevertheless, William Hedley's "Puffing Billy" must be regarded as one of the first practicable railway engines ever built. It is still to be seen in the South Kensington Museum, London. Patented in 1813, it began regular work at Wylam in that year, and continued in use until 1872. It was probably this engine which Stephenson saw when he said to Jonathan Foster that he could make a better, and it was no doubt the first to work by smooth wheels on smooth rails. Altogether it has been looked upon as the "father" of the enormous number of locomotives which have followed.

Mr. Blackett was a friend of Richard Trevithick; and among the various inventors and improvers of the locomotive engine Richard Trevithick, a tin-miner in Cornwall, must have a high place.

Trevithick was a pupil of Murdock, who was assistant of James Watt. Murdock had made a model successfully of a locomotive engine at Redruth. Others also had attempted the same thing. Savery had suggested something of the kind; Cugnot, a French engineer, built one in Paris about 1763; Oliver Evans, an American, made a steam carriage in 1772; William Symington, who did so much for the steamboat, constructed a model of one in 1784. So that many minds had been at work on the problem.

But Richard Trevithick was really the first Englishman who used a steam-engine on a railway. He had not much money and he persuaded his cousin, Andrew Vivian, to join him in the enterprise. In 1802 they took out a patent for a steam-engine to propel carriages.

But before this he had made a locomotive to travel along roads, and on Christmas Eve, 1801, the wonderful sight could have been seen of this machine carrying passengers for the first time. It is indeed believed to have been the first occasion on which passengers were conveyed by the agency of steam—the pioneer indeed of a mighty traffic.

The machine was taken to London and exhibited in certain streets, and at length, in 1808, it was shown on ground where now, curiously enough, the Euston Station of the London and North-Western Railway stands. Did any prevision of the extraordinary success of the locomotive flash across the engineer's brain? Before the infant century had run its course what wonderful developments of the strange new machine were to be seen on that very spot!

Much interest was aroused by the exhibition of this machine, and Sir Humphrey Davy, a fellow Cornishman, is reported to have written to a friend—"I shall soon hope to hear that the roads of England are the haunts of Captain Trevithick's dragons—a characteristic name."

His letter tends to show that the idea then was that the engine should run on the public roads, and not on a specially prepared track like a railway. Had not this idea been modified, and the principle of a railroad adopted, it is hardly too much to say that the extraordinary development of the locomotive would not have followed.

Trevithick's first engine appears to have burst. At all events, in the year 1803 or 1804, he built, and began to run, a locomotive on a horse tramway in South Wales. It appears that he had been employed to build a forge-engine here, and thus the opportunity was presented for the trial of a machine to haul along minerals. This, it is believed, was the first railway locomotive, and its builder was Richard Trevithick.

The trial, however, was not very successful. Trevithick's engine was too heavy for the tramway on which it ran, and the proprietors were not prepared to put down a stronger road. Furthermore, it once alarmed the good folk, unused then to railway accidents, by actually running off its rail, though only travelling at about four or five miles an hour. It had to be ignominiously brought home by horses. That settled the matter. It became a pumping engine, and as such answered very well.

In this locomotive, however, it should be noted Trevithick employed a device which, a quarter of a century later, Stephenson made so valuable that we might call it the very life-blood of the Locomotive. We mean the device of turning the waste steam into the funnel (after it has done its work by driving the piston), and thus forcing a furnace draught and increasing the fire. Stephenson, however, sent the steam through a small nozzled pipe which made of it a veritable steam-blast, while Trevithick, apparently, simply discharged the steam into the chimney.

Disgusted it would seem by the failure, the inventor turned his attention to other things. Trevithick appears to have lingered on the very brink of success, and then turned aside. Another effort and he might have burst the barrier. But it was not to be; though if any one man deserve the title, Inventor of the Locomotive, that man is the Cornish genius Trevithick. Readers who may desire fuller information of Trevithick and his inventions will find it in his "Life" by Francis Trevithick, C.E., published in 1872.

It must be borne in mind that Stephenson found the imaginary hindrance that smooth wheels would not grip smooth rails, cleared away for him by Hedley's experiment, whereas Trevithick had to contend against this difficulty. He strove to conquer it by roughing the circumference of his wheels by projecting bolts, so that

they might grip in that way. That is, his patent provided for it, if he did not actually carry out the plan.

It is very significant that this imaginary fear should have hindered the development of the locomotive. The idea seems to have prevailed that, no matter how powerful the engine, it could not haul along very heavy loads unless special provision were made for its “bite” or grip of the rails. Another difficulty with which Trevithick had to contend was one of cost. It is said that one of his experiments failed in London for that reason. This was apparently the locomotive for roads, as distinct from the locomotive for rails. A machine may be an academic triumph, but the question of cost must be met if the machine is to become a commercial and industrial success.

Mr. Blenkinsop of Leeds then took out his patent in 1811 for a rack-work rail and cogged wheel; but before this Mr. Blackett of Wylam had obtained a plan of Trevithick’s engine and had one constructed. He had met Trevithick at London, and it was as early as 1804 that he obtained the plan. The engines, therefore, of Mr. Blackett which Stephenson saw, came, so to speak, in direct line from Trevithick, except that Mr. Blackett’s second engine was a combination of Blenkinsop’s and Trevithick’s.

Some progress was made, but when on that memorable day George Stephenson, the engine-wright of Killingworth, said, “I think I could build a better engine than that,” no very effective or economical working locomotive was in existence.

Back therefore went George Stephenson to his home. He had seen what others had done, and with his knowledge of machinery and his love for engine work he would now try what he could do.

Would he succeed?

CHAPTER II.

GLANCING BACKWARDS AND STRUGGLING FORWARDS.

“My lord, will you spend the money to build a Travelling Engine?”

“Why? what would it do?”

“Haul coals to the Tyne, my lord. The present system of hauling by horses is very costly.”

“It is. But how would you manage it by a Travelling Engine?” Thereupon George Stephenson the engine-wright proceeded to explain.

In some such manner as this we can imagine that Stephenson opened up the subject to Lord Ravensworth, the chief partner in the Killingworth Colliery; and he won his lordship over.

Stephenson had already improved the colliery engines, and Lord Ravensworth had formed a high opinion of his abilities. So after consideration he gave the required consent.

Now, let us endeavour to imagine the position. The steam engine, of which the locomotive is one form, had been invented years before. The Marquis of Worcester made something of a steam engine which apparently was working at Vauxhall, South-west London, in 1656. It is said that he raised water forty feet, and by this we may infer that his apparatus was a steam-pump. He describes it in his work “Century of Inventions,” about 1655, and he is generally accredited with being the inventor of the steam engine. It was, however, a very primitive affair, the boiler being the same vessel as that in which the steam accomplished its work.

Captain Savery took the next step. He was the first to obtain a patent for applying steam power to machinery. This was in 1698, and he used a boiler distinct from the vessel where the steam was to exert its power. Savery’s engines appear to have been used to drain mines.

His engines acted in this way—the steam was condensed in a vessel and produced a vacuum which raised the water; then the steam pressing upon it raised it further in another receptacle.

An obvious improvement was the introduction of the piston. This was Papin’s idea, and he used it first in 1690. Six years later an engine was constructed by Savery, Newcomen (a Devonshire man), and Cawley, in which the “beam” was introduced, and also the ideas of a distinct boiler separate from a cylinder in which worked a piston. This machine was in operation for about seventy years. The beam worked on an axle in its centre—something like a child’s “see-saw,” and one end being attached to the piston moving in the cylinder, it was worked up and down, the other end of the beam being fastened to the pump-rod, which was thus alternately raised and depressed.

The upward movement of the piston having been effected by a rush of steam from the boiler upon its head, the steam was cut off and cold water run in upon it from a cistern. The steam was thus condensed by the water and a vacuum caused, and the piston was pressed down by the weight of the atmosphere—of course dragging down its end of the beam, and raising the pump-rod. The steam was then turned on again and pushed up the piston, and consequently the end of the beam also. Thus the engine continued to work, the turning of the cocks to admit steam and water being performed by an attendant. The engine was, however, made self-acting in this respect, and Smeaton improved this form of engine greatly. The beam is still used in engines for pumping.

Nevertheless, improved though it became, it was still clumsy and almost impracticable. It was the genius of James Watt which changed it from a slow, awkward, cumbrous affair into a most powerful, practicable, and useful machine.

His great improvements briefly were these: he condensed the steam in a separate vessel from the cylinder, and thus avoided cooling it and the consequent loss of steam power; secondly, he used the steam to push back the piston as well as to push it forward (this is called the “double-acting engine,” and is now always used); thirdly, he introduced the principle of using the steam expansively, causing economy in working; and fourthly, he enabled a change to be made of the up and down motion of

the piston into a circular motion by the introduction of the crank.

[Illustration: JAMES WATT.]

The use of the steam expansively is to stop its rush to the cylinder when the piston has only partially accomplished its stroke, leaving the remainder of the stroke to be driven by the expansion of the steam. In early engines the steam was admitted by conical valves, worked by a rod from the beam. Murdock, we may add in parenthesis, is believed to have invented the slide-valve which came into use as locomotives were introduced, and of which there are now numerous forms. The valve is usually worked by an “eccentric” rod on the shaft of the engine.

Watt was the author of many other inventions and improvements of the steam engine. Indeed, although Savery and Newcomen and others are entitled to great praise, it was Watt who gave it life, so to speak, and made it, in principle and essence, very much that which we now possess. There have, indeed, been improvements as to the boiler, as to expansive working, and in various details, since his day; but, apart from the distinctive forms of the locomotive and the marine engine, the machine as a whole is in principle much as Watt left it.

The centre of all things in a steam engine is usually the cylinder. Here the piston is moved backward and forward, and thence gives motion as required to other parts of the machine.

The cylinder is in fact an air-tight, round box, fitted with a close-fitting, round plate of metal, to which is fixed the piston-rod. Now, it must be obvious that if the steam be admitted at one end of the cylinder it will, as it rushes in, push the metal plate and the piston outward, and if this steam be cut off, and the steam admitted to the other end of the cylinder, it will push the metal plate and piston back again.

But what is to be done with the steam after it has accomplished its work? It may be permitted to spurt out into the air, or into a separate vessel, where it may be condensed. In the locomotive, under Stephenson’s able handling, this escape of steam was created into a steam-blast in the chimney to stimulate the fire. In compound and triple-expansion engines the steam is used—or expanded, it is called—in two or three cylinders respectively. When steam is condensed, it may be returned to the boiler as water.

It was the repairing of a Newcomen engine that seems to have started Watt on his inventions and improvements of the steam engine. He was then a mathematical instrument maker at Glasgow. As a boy he had suffered from poor health, but had been very observant and studious; and it is said that his aunt chided him on one occasion for wasting time in playing with her tea-kettle. He would watch the steam jetting from its spout, and would count the water-drops into which the steam would condense when he held a cup over the white cloud.

Delicate though he was in health, he studied much, and came, indeed, to make many other articles besides mathematical instruments. When, therefore, the Newcomen engine needed repair, it was not unnatural that it should be brought to him. It appears to have been a working model used at Glasgow University. He soon

repaired the machine; but, in examining it, he became possessed with the idea that it was very defective, and he pondered long over the problem—How it might be improved. What was wanting in it? How could the steam be condensed without cooling the cylinder?

Suddenly, one day, so the story goes, the idea struck him, when loitering across the common with bent brows, that if steam were elastic, it would spurt into any vessel empty of air. Impatiently, he hastened home to try the experiment. He connected the cylinder of an engine with a separate vessel, in which the air was exhausted, and found that his idea was correct; the steam did rush into it. Consequently the steam could be condensed in a separate vessel, and the heat of the cylinder maintained and the loss of power prevented. This invention seems simple enough; yet it increased the power of an engine threefold, and is at the root of Watt's fame. We must remember that the inventions which in process of time may appear the simplest and the most commonplace, may be the most difficult to originate. And it may fairly be urged—If it were so very simple, and so very obvious, why was it not invented before? The supposition is that in those days it was not so simple. It is possible that the great elasticity of steam was not sufficiently understood. In any case, the discovery and its application are regarded as his greatest invention.

Yet ten years elapsed before he constructed a real working steam engine, and so great we may suppose were the difficulties he encountered, including poorness of health, that once he is reported to have exclaimed: "Of all things in the world, there is nothing so foolish as inventing."

But a brilliant triumph succeeded. Eventually Watt became partner with Mr. Matthew Boulton, and the firm of Boulton & Watt manufactured the engine at Soho Ironworks, Birmingham. Mining proprietors soon discovered the value of the new machine, and Newcomen's engine was superseded for pumping.

Watt continued to improve the machine, and together with Boulton also greatly improved the workmanship of constructing engines and machinery. In a patent taken out in 1784, he "described a steam locomotive"; but for some reason he did not prosecute the idea. It is possible that the notion of building a special road for it to run upon did not occur to him, or appear very practicable.

His work was done, and it was a great work; but it was left for others to develop the steam engine into forms for hauling carriages on land or propelling ships upon the sea. Trevithick, Stephenson, and others did the one; Symington, Bell, and others did the second. Watt died in 1819, and though so delicate in youth, he lived to his eighty-fourth year.

The steam engine, therefore, as Watt left it, was practically as Stephenson came to know it. He would be acquainted with it chiefly as a pumping machine. But he saw what others had done to adopt it as a locomotive, and he now set to work.

Stephenson's first engine did not differ very materially from some of those which had preceded it. He was, so to speak, feeling his way. The machine had a round, wrought-iron boiler, eight feet long, with two upright cylinders placed on the top of it. At the end of the pistons from the cylinders were cross-rods connected with cogged

wheels below by other rods. These cogged wheels gave motion to the wheels running on the rails by cogs not very far from the axles. Stephenson abandoned the cogged rail, and adopted smooth wheels and smooth rails; but he did not connect the driving-wheel direct with the piston, the intervening cogged wheels being thought necessary to unite the power of the two cylinders.

In adopting the principle of smooth wheels on smooth rails, it is said that Stephenson proved by experiment that the arrangement would work satisfactorily. Mr. Smiles writes that Robert Stephenson informed him, “That his father caused a number of workmen to mount upon the wheels of a waggon moderately loaded, and throw their entire weight upon the spokes on one side, when he found that the waggon could thus be easily propelled forward without the wheels slipping. This, together with other experiments, satisfied him of the expediency of adopting smooth wheels on his engine, and it was so finished accordingly.” Thus it may be said that this obstacle—imaginary though it largely proved to be—was cleared away from Stephenson’s first engine.

Ten months were occupied in building the machine, and at last came the day of its trial. This was the 25th of July, 1814. Would it work?

Jolting and jerking along, it did work, hauling eight carriages at a speed of about four or six miles an hour—as fast as a brisk man could walk. Then came the question—Would it prove more economical than horse-power?

Calculations therefore were made, and after a time it was found that “Blucher” as the engine was called, though we believe its real name was “My Lord,” was about as expensive as horse-power.

The locomotive needed something more, some magic touch to render it less clumsy and more effective. What was it?

Then came the first great practicable improvement after the smooth wheels on smooth rails. It was the steam-blast in the funnel, by which the draught in the furnace was greatly increased. Indeed, the faster the engine ran the more furiously the fire would burn, the more rapid would be the production of steam, and the greater the power of the engine.

At first Stephenson had allowed his waste steam from the cylinders to blow off into the air. So great was the nuisance caused by this arrangement that a law-suit was threatened if it were not abated.

What was to be done with that troublesome waste steam? Now, whether Stephenson originated the idea or adapted what Trevithick had done, we cannot say, but at all events he achieved the object, wherever he gained the idea. He turned his exhaust steam through a pipe into the funnel, and at a stroke increased the power of his engine two-fold.

But that expedient was not alone. Stephenson had watched the working of “Blucher” to some purpose, and he decided to build another engine with improvements.

The cumbersome cog-wheels must go; they complicated the machine terribly, and prevented its practicability. Therefore in his second engine he introduced direct connection between the pistons and the wheels. There were a couple of upright cylinders as before, with cross-rods attached to the piston-ends, and connecting rods from the end of each cross-rod, reaching down to the wheels. But to overcome the difficulty of one wheel being at some time higher than the other on the poorly constructed railway of that period, a joint was introduced in the cross-rod, so that if, perchance, the two wheels should not be always on exactly the same level, no undue strain should be placed on the cross-rod. Furthermore, the two pairs of wheels were combined first by a chain, but afterwards by connecting rods. This may be called the locomotive of 1815, the year in which the patent was taken out.

[Illustration: EDWARD PEASE.]

The engine accomplished its work more satisfactorily than before, and was placed daily on the rails to haul coal from the mine to the shipping point. But still its economy over horse-power was not so great as to cause its wide adoption. And it was still little better, if anything, than a mere coal haul.

Nevertheless Stephenson persevered. He was appointed engineer to the Stockton and Darlington Railway—an enterprise largely promoted by Mr. Edward Pease. It was opened on the 27th of September, 1825, and a local paper writes as follows:—

“The signal being given, the engine started off with this immense train of carriages, and such was its velocity, that in some parts the speed was frequently 12 miles an hour; and at that time the number of passengers was counted to be 450, which, together with the coals, merchandise, and carriages, would amount to near 90 tons. The engine, with its load, arrived at Darlington, a distance of 8¾ miles, in 65 minutes. The 6 waggons loaded with coals, intended for Darlington, were then left behind; and obtaining a fresh supply of water, and arranging the procession to accommodate a band of music and numerous passengers from Darlington, the engine set off again, and arrived at Stockton in 3 hours and 7 minutes, including stoppages, the distance being nearly 12 miles.”

Stephenson became a partner in a business for constructing locomotives at Newcastle, and three engines were made for the Stockton and Darlington Railway. Nevertheless they appear to have been used chiefly if not almost entirely for hauling coal; for the passenger-coach called the _Experiment_ was hauled by a horse, and the journey occupied about two hours.

The locomotive was not even yet a brilliant success over horse-power. What was to be the next step?

CHAPTER III.

FIFTEEN MILES AN HOUR.

Five hundred pounds for the best locomotive engine!

So ran the announcement one day in the year 1829. The Liverpool and Manchester Railway was nearly completed, but yet the directors had not fully decided what power they would employ to haul along their waggons.

Horse-power had at length been finally abandoned, and numbers of schemes had been poured in upon the managers. But the contest seemed at last to resolve itself chiefly into a rivalry between fixed and locomotive engines. Principally, if not entirely, swayed however by the arguments of George Stephenson, the directors yielded to the hint of a Mr. Harrison, and offered a £500 prize.

The engine was to satisfy certain conditions. Its weight was not to be above six tons; it was to burn its own smoke, haul twenty tons at a rate of ten miles an hour, be furnished with two safety valves, rest on springs and on six wheels, while its steam pressure must not be more than fifty lbs. to the square inch. The cost was not to exceed £550.

Stephenson, who was the engineer of the Railway, decided to compete. He was now in a very different position from that which he occupied when he built his second locomotive in 1815. His appointment as engineer to the Stockton and Darlington Railway had greatly aided his advancement, and when it was decided to build a railway between the two busy cities of Manchester and Liverpool it was not unnatural that he should take part in the undertaking.

The idea of constructing rail, or tram ways, was not new. Railways of some kind were used in England about two hundred years before, that is, about the beginning of the seventeenth century. Thus Roger North writes:—“The manner of the carriage is by laying rails of timber from the colliery to the river, exactly straight and parallel; and bulky carts are made with four rollers fitting those rails, whereby the carriage is so easy that one horse will draw down four or five chaldron of coals, and is an immense benefit to the coal merchants.”

It is said that the word tramway is derived from tram, which was wont to mean a beam of timber and also a waggon. In any case, such rough ways were introduced in mining districts, for, as may be readily believed, one horse could draw twenty times the load upon them that it could on an ordinary road.

The old ways were first made of wood, then of wood faced with iron, then altogether of iron.

Now, in making his railway between Liverpool and Manchester, Stephenson had many difficulties to encounter. He decided that the line should be as direct as possible. But to accomplish this, he would have to pierce hills, build embankments, raise viaducts, and, hardest of all, construct a firm causeway across a treacherous bog called Chat Moss.

“He will never do it,” said some of the most famous engineers of the day. “It is impossible!”

Impossible it certainly seemed to be. Chat Moss was like a sponge, and how was an engineer to build a solid road for heavy trains over four miles of soppy sponge! A person could not trust himself upon it in safety, and when men did venture, they

fastened flat boards to their feet, something after the fashion of snow-shoes, and floundered along upon them.

Stephenson began by taking the levels of the Moss in a similar manner. Boards were placed upon the spongy moss, and a footpath of heather followed. Then came a temporary railroad. On this ran the trucks containing the material for a permanent path, which were pushed by boys who learned to trot along easily on the narrow rails.

Drains were dug on either side of the proposed road, and tar-barrels covered with clay were fitted into a sewer underneath the line in the middle of the Moss. Heather, hurdles, tree branches, etc., were spread on the surface, and in some parts an embankment of dry moss itself was laid down. Ton after ton of it disappeared until the directors became alarmed, and the desperate expedient of abandoning the works was considered.

But Stephenson was an Englishman out and out. He never knew when he was beaten. “Keep on filling,” he ordered; and in spite of all criticism and all alarm, he kept his hundreds of navvies hard at work, pouring in load after load of dry turf.

It must be borne in mind, however, that Stephenson did not continue blindly at his task. He had good reason for what he did. His persistence was a patient, intelligent perseverance, and not a stupid obstinacy. His main arguments seem to have been two. He judged that if he constructed a sufficiently wide road, it would float on the moss, even as ice or a raft of wood floats on water and bears heavy weights; and secondly, he seems to have been animated by the idea, that, if necessary, he could pour in enough solid or fairly solid stuff to reach the bottom and rise up to the surface in a hard mass.

Both ideas seem to have been realised in different parts of the bog. Joy took the place of despair, and triumph exulted over discouragement, as at length the solid mass appeared through the surface. Furthermore, the expense was found to be none so costly after all. No doubt any quantity of turf could be obtained from the surrounding parts of the Moss and dried.

At another part of the railway called Parr Moss an embankment about a mile and a-half was formed by pouring into it stone and clay from a “cutting” in the neighbourhood. In some places twenty-five feet of earth was thus concealed beneath the Moss. The eye of the engineer had as it were pierced through the bog and seen that his solid bank was steadily being built up there.

Before, however, the road across Chat Moss was fairly opened, the trial of locomotives for the prize of £500 had taken place. The fateful day was the 1st day of October, 1829, and the competition was held at Rainhill. A grand stand was erected, and the side of the railway was crowded. Thousands of spectators were present. The future of the locomotive was to be decided on this momentous occasion.

Now, hitherto the difficulty in the locomotive had been to supply a steady and sufficient supply of steam to work the engine quickly and attain high speed and power. Partly, this had been accomplished by Stephenson’s device of the steam-blast in the funnel. But something more was needed.

That requirement was found in the tubular boiler. If the long locomotive boiler were pierced with tubes from end to end, it is clear that the amount of heating surface offered to the action of the fire would be greatly increased. It was this idea which was utilised in the “Rocket,” the engine with which Stephenson competed at Rainhill, and utilised more perfectly than ever before.

Trevithick himself seems to have invented something of the kind, and M. Seguin, the engineer of the St. Etienne and Lyons Railway utilised a similar method. But Henry Booth, the secretary of the railway which Stephenson was then building, invented a tubular boiler without, it is said, knowing anything of Seguin’s plan, and Stephenson who had already experimented in the same direction, adopted Booth’s method.

At first it was a failure. The boiler, fitted with tubes through which the hot air could pass, leaked disastrously, and Stephenson’s son, Robert, wrote to his father in despair. But again George said “persevere,” and he suggested a plan for conquering the difficulty. Again, it was a simple, but as the event proved, an effective plan.

The copper tubes were merely to be fitted tightly to holes bored in the boiler and soldered in. The heat caused the copper to expand and the result was a very strong and water-tight boiler. There were twenty-five of these tubes, each three inches in diameter, and placed in the lower portion of the boiler, leading from the furnace to the funnel. Water also surrounded the furnace. Further, the nozzles of the steam-blast pipes were contracted so as to increase the power of the blast, and consequently raise the strength of the draught to the fire.

[Illustration: “THE ROCKET.”]

The cylinders were not placed at the top of the boiler, but at the sides in a slanting direction, one end being about level with the boiler roof. They occupied a position mid-way between the old situation upright on the roof and their present position below, or at the lower portion. The pistons acted directly on the driving wheels by means of a connecting rod, and the entire weight of the engine with water supply was but 4½ tons.

On the day of trial only four engines competed. Many had been constructed, but either were not completed in time, or for various reasons could not be exhibited. The famous four were:—The “Novelty” by Messrs. Braithwaite and Ericsson; The “Rocket” by Messrs. R. Stephenson & Co.; The “Perseverance” by Mr. Burstall; and The “Sanspareil” by Mr. Timothy Hackworth. Each engine seems to have run separately, and the length of the course was two miles. The test was that the engine should run thirty miles, backwards and forwards, on the two mile level course, at not less than ten miles an hour, dragging three times its own weight.

The “Novelty” at first appears to have beaten the “Rocket,” for she ran at times at the rate of twenty-four miles an hour; while the first trip of the “Rocket” covered a dozen miles in fifty-three minutes. The engineers of the “Novelty” used bellows to force the fire, but on the second day these bellows gave way, and the engine could not do its work. The boiler of the “Sanspareil” also showed defects, but Stephenson’s “Rocket” calmly stood the strain. Practicable as usual, Stephenson’s work was as

good in its results, nay, even better than before, for he hooked the “Rocket” to a carriage load of thirty people, and rushed them along at the then surprising speed of between twenty-four to thirty miles an hour. Mr. Burstall’s “Perseverance” could not cover more than six miles an hour.

The competitions continued, but the “Novelty,” although running at the rate of twenty-four and even twenty-eight miles an hour, broke down again and yet again; its boiler plates appear to have gone wrong on one occasion; while the “Sanspareil” also failed, and furthermore blew a good deal of its fuel into the air because of the arrangement of its steam-blast.

But the more the “Rocket” was tried, the more practicable and reliable the engine appeared to be. On the 8th of October it gained a speed of 29 miles an hour, its steam pressure being about 50 lbs. to the square inch, and its average speed was fifteen miles an hour—that is, five miles an hour over the conditions required. These results appear to have been accomplished with a weight of waggons of thirteen tons behind it. When detached it ran at the rate of thirty-five miles an hour.

In short, the “Rocket” was the only locomotive which fulfilled all the conditions specified for the competition, and the prize was duly awarded to Stephenson and Booth.

The battle of the locomotive was won. Men could see that the machine was feasible and practicable; that it was a new force with immense possibilities before it.

How have those possibilities been realised?

CHAPTER IV.

A MARVEL OF MECHANISM.

“The time is coming when it will be cheaper for a working man to travel on a railway than to walk on foot.”

So prophesied George Stephenson some few years before his successful competition at Rainhill; and by his success on that fateful day, he had brought the time appreciably nearer. The directors of the Liverpool and Manchester Railway no longer debated as to what form of traction they should adopt.

But Stephenson did not rest on his laurels. Every new engine showed some improvement. The “Arrow” sped over Chat Moss at about 27 miles an hour, on the occasion of the first complete journey along the line, on the 14th of June, 1830; and when, on the public opening of the railway on the 15th of September, 1830, Mr. William Huskisson, M.P., was unhappily knocked down by the “Rocket,” George Stephenson himself took the maimed body in the “Northumbrian,” fifteen miles in twenty-five minutes—that is, he drove the engine at the speed of thirty-six miles an hour.

The sad death of Mr. Huskisson has often been referred to, but we may tell the story again, following the account given by Mr. Smiles, who had the advantage of the assistance of Robert Stephenson in the preparation of his biography.

The engines it appears halted at Parkside, some seventeen miles from Liverpool, to obtain water. The “Northumbrian,” with a carriage containing the Duke of Wellington and some friends, stood on one line, so that all the trains might pass him in review on the other. Mr. Huskisson had descended from the carriage and was standing on the rail on which the “Rocket” was rapidly approaching. There had been some coolness between the Duke and Mr. Huskisson, but at this time the Duke extended his hand and Mr. Huskisson hurried to grasp it, when the bystanders cried “Get in! get in.”

Mr. Huskisson became flurried and endeavoured to go round the carriage door which was open and hung over the rail; but while doing this, the “Rocket” struck him and he fell, his leg being doubled over the rail and immediately crushed. Unfortunately he died that evening at Eccles Parsonage.

This sad event cast a gloom over the otherwise rejoicing day; but the wonderful speed at which the wounded man was conveyed, proved a marvellous object lesson as to what the locomotive could accomplish.

In the “Planet,” put upon the line shortly after the opening, the cylinders were placed horizontally and within the fire box. The engine drew eighty tons from Liverpool to Manchester against a strong wind in two and a-half hours, while on another occasion with a company of voters, it sped from Manchester to Liverpool, thirty-one miles, in an hour. But next year the “Samson,” which was still further improved, and the wheels of which were coupled so as to secure greater grip on the rails, hauled 150 tons at twenty miles an hour with a smaller consumption of fuel.

The locomotive had now become one of the wonders of the world. Since then its speed has been doubled. But all the improvements (with possibly one exception—that of the compound cylinder which is at present only partially in use) have been more in details than in principles. Thus the 70 or 80 ton express engine, which covers mile after mile at the rate of a mile a minute without a wheeze or a groan, is not very different essentially from George Stephenson’s locomotives, though its steam pressure is very much higher.

There are, for instance, the multitubular boiler, the furnace surrounded by water and communicating with the boiler, the horizontal cylinders acting directly on the driving wheels, and the steam-blast by which the waste steam is spouted up the chimney, creating a draught in the furnace.

These may be regarded as the more important of the essential principles, although there is diversity of details, more especially for the different work required. But the steam pressure is now much greater. Let us glance at a typical English locomotive. You might not think it, but the machine has about five thousand different parts, all put together as Robert Stephenson said “as carefully as a watch.”

At first sight you will probably not see the cylinders. The tendency in many engines now seems to be to place them inside the wheels, for it is urged that the placing of the heavier parts of the mechanism near to the centre lessens oscillation, and protects the machinery more effectually. Against this, it is said that the placing of the cylinders in that position increases the cost and the complication of the driving axle, and renders the pistons and valves more inaccessible for the purposes of repair. Both

forms have their advocates, and the outside-cylinder form may be seen on the London and South-Western and some other railways, while the inside may be seen on the North-Western and others.

The boiler is of course the long, round body of the locomotive, and in English machines it is placed on a strong plate frame. Then as to the driving-wheels. Express engines, such as the splendid “eight-feet singles” of the Great Northern, have often, as the name implies, but one large driving-wheel on either side, and for great speeds this form is held to possess certain advantages. Certainly the performances of Mr. Patrick Stirling’s expresses would indicate that this is the case.

With steam raising the safety valve at a pressure of 140 lbs. to the square inch, the engines will whisk a score of carriages out of King’s Cross up the northern height of London at forty miles an hour, and then without a stop rush on to Grantham at near sixty. Standing on the platform at King’s Cross, with a large part of the immense driving-wheel hidden below you as it rests on the rail, you do not realise its tremendous size. Yet, let the engine-driver open the throttle, as it is called—that is, turn on the steam to the cylinders—and that huge wheel will revolve, and with its neighbour on the other side, haul after them that heavy train of carriages, and, gathering speed as they go, they will soon be rushing up the incline at forty miles an hour, and then on at sixty. It is a marvel of mechanism!

But then the compound engines that Mr. F. W. Webb, the engineer of the North-Western, builds for that Company can also perform remarkable things. The compound is the great modern improvement (some engineers might doubt whether improvement be the correct word) in the locomotive, effecting, it is said, an economy of from ten to fifteen per cent. in fuel. Now the compounding principle has been developed to such an extent in marine steam engines that it revolutionised steam navigation. But the application of the principle has not been so great in the case of the locomotive.

Briefly, the principle is this—the steam is sent out from the boiler at a high pressure, say 160 to 180 lbs. to the square inch, and is used in one or in a pair of high-pressure cylinders, and then used again, by means of its expanding power, in a larger, low-pressure cylinder. Mr. John Nicholson, of the Great Eastern Railway, suggested a compound locomotive before even the compound marine engine had been made, and his design was successful; but in 1881 Mr. Webb, of the North-Western, patented a compound locomotive, with two small high-pressure, and one large low-pressure cylinders, the latter twenty-six inches in diameter. Placed between the front wheels, the bright boss of this cylinder may be seen in shining steel as it flies over the rails.

The argument is that the compound burns less fuel and is more powerful than a non-compound of the same weight; but against this is launched the objection that the compound is more expensive to build, to repair, and to maintain. Still further it is argued, that a fast-speeding locomotive has not the time in its hurrying life to expand its steam in the tick of time between each stroke of the piston.

[Illustration:

THE COMPOUND LOCOMOTIVE “GREATER BRITAIN.” _By kind permission of Mr. F. W. Webb, L. & N. W. Railway._]

Mr. Worsdell’s compounds on the North-Eastern Railway have but two cylinders, one high and the other low-pressure. The one is eighteen and the other twenty-six inches across. Instead of the steam alternating between the two cylinders, it all passes first to the high-pressure and then, through a pipe in the smoke-box, to the larger low-pressure cylinder. These locomotives, it is said, are not under the objection alleged against the other compounds—viz., that they have more parts, and are more costly to build and maintain. Yet it is claimed for them that they are more economical and more powerful than non-compounds.

When doctors disagree who shall decide? The cost or speed might decide; but at present it seems doubtful on which side the balance does really fall. Engines of the three types have done splendid work. A Worsdell compound, built by Mr. Worsdell, of the North-Eastern Railway, is reported to have rushed down the incline to Berwick one day at seventy-six miles an hour for some miles at a time. Then the “Greater Britain,” a massive North-Western compound engine, turned out at the Crewe works in 1891, and weighing seventy-five tons, can whirl along with ease a heavy twenty-five coach express at an average of over fifty miles an hour, with a comparatively small consumption of fuel.

This locomotive was described in the _Engineer_ newspaper as the most remarkable that had been built in England for several years. Its axle bearings are of great length, and its parts are very substantial, so that it ought to keep out of the repairing shops for long spells of time. It was specially planned for both fast and heavy passenger traffic to Scotland, and its work on its trial trip was so good that it was confidently expected it would answer expectations. In working, the engine has been found to develop great speed and power, easily running at over fifty miles an hour with what is called a double train—viz., twenty-five coaches, behind it. Indeed, it has run at fifty-five miles with this heavy train. Its stated speed ranges from thirty to fifty-five miles an hour, with a low consumption of fuel.

This last is a matter of very great importance to engineers and railway directors; and when we state that, according to Mr. Bowen Cooke, the North-Western engines altogether burn 3095 tons of coal per day, any small saving per hour would be eagerly welcomed.

Now, it is claimed that the compounds have consumed about six pounds of coal per mile less than others on the same work, and that they also haul along loads which would require two of the other type. If so, the saving in the North-Western coal-bill must be enormous.

[Illustration: BACK AND FRONT VIEW OF THE LOCOMOTIVE “GREATER BRITAIN.”]

A great feature in this engine is a combustion chamber placed within the barrel of the boiler. This chamber catches all the gases from the furnace, and causes the heat generated by them to be used to the utmost for the production of steam. Though heavier than any engine previously built, yet it is so made that no greater weight than

usual rests upon any of the wheels, thus throwing no extra strain on the railway or the bridges. The two couples of driving-wheels are placed before the furnace, and an additional couple of small wheels behind the furnace, and beneath the foot-plate where the driver and fireman stand. The weight therefore is evenly distributed, with another pair of wheels to bear the burden. The front wheels are fitted with the radial axle-box patented by Mr. Webb, so that, although the engine is of great length, yet it can speed round curves with perfect safety.

Yet this engine, though one of the most remarkable developments of the locomotive, is in essence and in principle but very like the “Rocket.” The difference lies in its innumerable details, exhibiting so much engineering skill and ingenuity, in the compound cylinders, in higher pressure steam, and in its marvellous power and speed combined.

On the other hand, the Great Northern runs daily from Grantham to London at fifty-three and fifty-four miles an hour average; while it was reported in the _Engineer_ of the 10th of March, 1888, that a Great Northern train from Manchester to London, when running from Grantham to London, covered one mile in forty-six seconds, that is, at the rate of seventy-eight and a-quarter miles an hour, and two miles following each other were run in forty-seven seconds each, that is, seventy-six miles an hour. We doubt, indeed, if any railway in the world can show regular faster daily running than some of the Great Northern expresses between London and Grantham. The average speed of their Manchester train over this ground is slightly over fifty-four miles an hour. Then there are the Great Western expresses, the “Dutchman” and the “Zulu,” at only slightly less speeds, to say nothing of the fine performances of the Midland. We may take it, therefore, that the compound locomotives, excellent as their work has been, have not really beaten their rivals in point of speed.

Compounds are used largely on the North-Western, the Great Eastern, and the North-Eastern, and should they prove to be really more economical in working, while maintaining at least equal power and speed with their rivals, we have no doubt but that they will prevail.

CHAPTER V.

A MILE A MINUTE.

“The express is to be quickened, my lord. Mr. Thompson, the general manager, has given instructions to that effect.”

So spoke the station master at Carlisle, on the 17th of March, 1894, to Lord Rosebery.

His lordship had very recently been appointed Prime Minister, and was on his way to Edinburgh to deliver a great public speech. The train, presumably, was late, or he, through stress of business probably, had left too little margin of time. However, by the instructions of Mr. Thompson, the general manager of the Caledonian Railway, the express was accelerated, and it rushed over 101 miles in 105 minutes, one of the

quickest locomotive runs, we imagine, that have ever been recorded. The train arrived fifteen minutes before it was due, and Lord Rosebery was enabled to keep his engagement.

This run was approximately at the rate of a mile a minute, and maintained for an hour and three-quarters. Only some two years or so previously a somewhat similar run was made. An officer of the Guards found that he had lost the south-going mail train at Stirling. He had been on leave in Scotland, and was bound to report himself in London next morning.

What was he to do? Did he sit down and moan, or fly to the telegraph office and endeavour to excuse himself? Not he. He promptly engaged a special train, which flying over the metals, actually caught the mail at Carlisle, having covered 118 miles in 126 minutes; that is, again, approximately a mile a minute, and maintained for slightly over two hours.

Now, in order to attain high average speed, some parts of the journey, say very easy inclines or levels, must be covered at a much higher rate. Thus, to obtain an average of fifty-two miles an hour—which is probably the regular average of our best English expresses—the pace will most likely be sometimes at the rate of seventy, or it may be seventy-six, miles per hour.

The United States have claimed to run the fastest regular train. This is the “Empire State Express” of the New York Central, which bursts away from New York to Buffalo, a trip of 140 miles, at the average rate of 52-12/100 miles per hour, but running eighty miles at the rate of 56¾ miles an hour. It is also said that, in August, 1891, a train on the New York portion of the Reading road ran a mile in less than forty seconds, and covered a dozen miles at an average of barely 43½ seconds per mile.

English expresses could certainly accomplish these average speeds, but the fact is very high speeds do not pay. They wear everything to pieces. Then there is the coal consumption. American railway engineers—according to the _Engineer_ newspaper—“seem to be unable to get on with less than 100 lbs. per square foot (of fire grate area) as a minimum;” while, from the same paper, we learn that the average rate of burning of Mr. Webb’s remarkable North-Western engine, the “Greater Britain,” was but “a little over seventy-three lbs. per square foot per hour,” or, altogether, 1500 lbs. per hour.

The rails also are greatly worn by continuous high speeds. Engineers have been equal to this difficulty, and rails are now made of steel, and even steel sleepers are constructed on which the rails repose. But still the wear and tear, especially to engines, of continuous high speeds, is very great. The reason why the famous “Race to Edinburgh” was stopped was doubtless because of the needless wear and tear. Surely an average of fifty to fifty-two miles an hour is fast enough for all ordinary purposes. If greater speed can be obtained without too great a cost, well and good; but if not, the public must be content.

Nevertheless, during that famous “Race” in the summer of 1888, some magnificent engine work was accomplished. Thus, for instance, the North-Western and their

partners actually ran from Euston to Edinburgh, 400 miles, in 427 minutes. Then the Great Northern and their partners, the East Coast route, next day covered 393 miles in 423 minutes, this journey including 124½ miles from Newcastle to Edinburgh covered in 123 minutes. This speed is, of course, more than a mile a minute, and kept up for slightly over two hours.

The third-class passenger was at the root of the matter. Companies are finding out they must consult his convenience; and the beginning of the "Race" was probably the announcement that the "Flying Scotchman"—the 10 o'clock morning train from King's Cross—would carry third-class passengers. Hitherto it had beaten its rival, the West Coast route (run by the North-Western and its partner, the Caledonian), as to speed, but had conveyed only first and second-class passengers.

Thereupon the West Coast announced that they would reach Edinburgh in nine hours. As this route is harder for engines—for it climbs the Cumbrian Hills, and is, moreover, seven miles longer—this would mean faster running and harder work than its rivals. The Great Northern, which according to its well-deserved reputation probably tops the world for speed, could not brook this, so the East Coast route reduced its time from nine hours to eight hours and a-half.

So the contest stood for about a month, when the West Coast calmly announced the same time for its journey. Thenceforward the blows fell thick and fast. It was a battle of giants, but fought with good temper and gentlemanly honour on both sides.

The West Coast were arriving at Edinburgh at half-past six. "The Flying Scotchman," by the East Coast route, thereupon drew up in the Scotch capital at six o'clock. Then the West Coast ran to Edinburgh in eight hours, stretching away from Euston to Crewe, 158½ miles in 178 minutes, without a stop—probably the longest run without a break ever made. The Caledonian Company, the North-Western's partner, then ran from Carlisle to Edinburgh, 100¾ miles, in 104 minutes. The North-Western thereupon actually ran from Preston to Carlisle, over the Cumberland Hills, ninety miles in ninety minutes—a magnificent performance hard indeed to beat, if, in fact, it ever has been really beaten; while, later on, the same Company ran from Euston to Crewe in 167 minutes instead of their remarkable 178 minutes a few days previously. This, with the other accelerations, gave the West Coast their record run of 400 miles in 427 minutes of running time, which took place on the 13th of August. But the East Coast had also accelerated, the North-Eastern covering 205 miles in 235 minutes, and the Great Northern rendering an equally good, if not better, performance, the whole 393 miles being covered in 423 minutes. Some of the miles on the East Coast route sped by at the rate of seventy-six an hour.

To accomplish these runs the weight of trains was cut down, and the times of stoppages reduced or abolished altogether. But the expense was too great. It did not really "pay" in convenience or in money, and to these judgments companies must bow. But considering that the Great Northern reaches Grantham, 105¼ miles, in 115 minutes as a daily occurrence, an approximate running of near a mile a minute, and that the North-Western can run at an average of fifty-five miles an hour, the locomotive has amply justified George Stephenson's prophecy when he made

“Blucher,” that there was no limit to the speed of the locomotive, provided the work could be made to stand.

Mr. C. R. Deacon also prophesied a few years since in an American magazine that a hundred miles an hour would be the express speed of the future, provided that passengers would give up luxurious cars and dining and sleeping carriages. At present it seems questionable if they will do so.

[Illustration: THE “FLYING DUTCHMAN.”]

But speed is by no means the monopoly of the North. Other companies beside the owners of the East and West Coast routes to Scotland can run expresses equally or almost as fast. There is the “Flying Dutchman,” for instance, of the Great Western. It daily covers the 77¼ miles from London to Swindon in 87 minutes. And the tale is told by Mr. W. M. Acworth, on the authority of an inspector who was in charge of the train, that a famous Great Western engine, the “Lord of the Isles,” which was in the Exhibition of 1851, actually whirled a train from Swindon to London, 77¼ miles in 72 minutes.

Some of those older engines could run bravely. Mr. Acworth reports that “a Bristol and Exeter tank-engine with 9 feet driving wheels, a long extinct species,” pelted down a steep incline at the speed of 80 miles an hour, many years since, and it has never been surpassed. The fastest speed during the Race to Edinburgh days seems to have been 76 miles, but perhaps the weight of the trains may have accounted for this. Mr. Acworth himself is believed to have accomplished the fastest bit of advertised journeying in the world. He went down on the “Dutchman,” and leaving Paddington at 11.46, he caught the return train at Swindon and was back at 2.45, having covered 154½ miles, with five minutes for refreshments, in 177 minutes. The line is easier on the up journey to London, and mile after mile sped by at a rate of over 60 miles an hour. From 56½ to 58 seconds was the chronograph’s record again and again, while on the down journey to Swindon he records a burst of 34½ miles in 34 minutes.

The gradients of the railway form of course a most important factor in the question of speed. The Midland has one of the hardest roads in England for steep slopes, yet its magnificent engines bring its heavy trains from Leicester, 99¾ miles in 122 minutes. Considering the high levels the locomotives have to climb, only to sink again to low flats, as about the Ouse at Bedford, this performance is really as fine as some of the superb running of the Great Northern.

The Southern lines out of London have no long distances to cover as the Northern, unless it may be the South-Western to Plymouth. The South-Western to Bournemouth and Exeter, and the mail trains on the South-Eastern, Chatham and Dover, and the Brighton trains can also show some excellent work as regards speed.

The government of a large railway now has grown to something like the rule of a small state. Sir George Findlay, the general manager of the North-Western Company, in his evidence before the Labour Commission in 1892, deposed that the capital raised for British railways amounted to the vast sum of 897 millions of pounds; that the receipts were 80 millions yearly, that much more than half of this immense amount, namely 43 millions, yearly was paid in wages, and that half-a-million of men

directly or indirectly were given employment.

To such enormous dimensions has the railway developed. And the locomotive engine is the centre and soul of it all. Stephenson got it, so to speak, on its right lines of working, and it has run along them ever since, until in its great capacity for speed, its power for drawing heavy loads, and its strength and beauty of construction it may fairly be called one of the wonders of the world.

[Illustration]

[Illustration]

THE STORY OF THE STEAMSHIP.

CHAPTER I.

THE “COMET” APPEARS.

“If only people could reach the place easier, I could do more business.”

So mused Henry Bell of Glasgow about the year 1810. He was an ingenious and enterprising man, and he had established a hotel or bathing-house at Helensburgh on the Clyde. But he wanted more visitors, and he puzzled his brain to discover how he could offer facilities for them to reach the place.

He tried boats, worked by paddles, propelled by hand; but these proved a failure. They had been in use years before, though perhaps he knew it not. Tradition says that boats fitted with paddle wheels and worked by oxen in the boat, were known to the Egyptians, but perhaps tradition is wrong. The Romans and the Chinese also are said to have known wheel boats, the wheels worked by men or by animals—in the case of the Chinese apparently by men alone. A similar kind of boat appears to have been tried on the Thames in the seventeenth century; but whether Bell knew of these things or not, his experiments of the same kind did not answer. What was to be done?

He determined to build a steamboat. At first sight there does not seem to be much connection between baths and steamboats, but apparently it was the ownership of the one which led Henry Bell to build the other, and to become the first man in Great Britain who used a steamboat for what may be called public and commercial purposes.

She was a queer craft. Her funnel was bent and was used also as a mast, and she poured forth quantities of thick smoke. But she was successful, and laboured along at the rate of five miles an hour. Up and down the river she plied, and whatever else she did, or did not, she made the good folk of those days understand that steam could be applied to navigation.

She was called the _Comet_, not because, even in the opinion of her owner, she resembled a blazing meteor, but because, to use Bell’s own words, “she was built and finished the same year that a comet appeared in the north-west part of Scotland.”

“Whatever made you think of starting a steamship?” we can imagine a friend asking him as they stood on the bank and watched the _Comet_ with her paddles

shaped like malt shovels, splashing up the water.

“Partly it was Miller’s experiments, and partly it was a letter from Fulton. You know, Fulton has put the _Clermont_ successfully on American waters. He had been over here talking with Symington, who had a steamer on the Forth and Clyde Canal you remember, and he wrote to me also asking about machinery and requesting me to inquire about Miller’s boats, and send him drawings.”

“And did you?”

“Oh ay, I did; but when he replied afterwards that he had made a steamboat from the drawings though requiring some improvements, I thought how absurd it was to send my opinions to other countries and not put them into practice in our own.”

“So you made the _Comet_?”

“Well, I made a number of models before I was satisfied; but when I was convinced the idea would work, I made a contract with John Wood & Co., of Port-Glasgow, and they built me this boat, which I fitted up with engine and paddles, as you see. John Robertson actually set up the engine. We will go aboard presently, and you shall see her.”

[Illustration: BELL’S “COMET.”]

They did so, and this is something of what they saw. They found a small vessel, forty feet long and ten and a-half wide, and only about twenty-five tons burthen. The furnace was bricked round, and the boiler, instead of being in the centre, was seated on one side of the ship, with the engine beside it. But the funnel was bent and rose aloft in the middle, and it answered the purpose of a mast—to carry sail.

“But look at the machinery,” we can imagine Bell saying to his friend. “We have one single cylinder, you see. The piston is attached to a crank on an axle. This axle carries a big cog wheel, which, working two more placed on the paddle axles, causes them to revolve.”

“And the paddles?”

“Well, you see, we have now two sets on each side, and each paddle is shaped something like a malt shovel; but I think I shall alter them, and have paddle wheels soon.”

Bell carried out his improvement, and in a short time he did adopt the better form of paddle wheel. The improved _Comet_, with a new engine, attained six or seven miles an hour. But before this, Mr. Hutchison, a brewer, built another boat, bigger than the _Comet_, and her engine was of ten horse-power, while the _Comet’s_ was but three. She travelled at an average of nine miles an hour, and her fares were but a-third of those charged by coach.

The news of the steamers on the Clyde became noised abroad, and steamboats began to appear on other British rivers. The success of the new venture became assured.

But how had it been brought about? Bell had referred to the labours of others, and, indeed, his was not the first steamboat, though, doubtless, it was the first in Britain to ply for passengers.

The truth is, that as with the locomotive, several minds were working towards the same object. And among those early steamboat seekers Patrick Miller, of Dalswinton, and William Symington, of Wanlockhead Mines, are entitled to high place.

Indeed, Symington is said to have built the “first practically successful steamboat” in the world. She was called the _Charlotte Dundas_, and, in 1802, she tugged two barges, together of about 140 tons, nineteen and a-half miles, in six hours, with a strong wind against her.

She was built under the patronage of Lord Dundas, and was intended to be used for towing on the Forth and Clyde Canal, but the proprietors of the canal would not adopt this new method of propulsion; they feared that the wash from the wheels would damage the canal banks. So the _Charlotte Dundas_, successful though she was to a certain extent, had to be beached and broken up. But Fulton and Bell both inspected her, and we may infer that what they saw, influenced their subsequent action.

The engine of the _Charlotte Dundas_ was of the “double action” character, introduced by Watt, and it turned a crank in the paddle wheel shaft. The wheel was placed at the stern; and boats with their wheels thus placed are still made for use in particular places. Thus Messrs. Yarrow built one in 1892, to voyage in the shallow rivers and lagoons on the west coast of Africa; the idea being that a screw-propeller would have been likely to become fouled with weeds.

The _Charlotte Dundas_, we say, has been regarded as the “first practically successful steamboat ever built.” No doubt it was so, and the credit must be largely given to William Symington. But his success, and that which crowned the labours of others, were rendered possible by the inventions and improvements of James Watt.

Others had experimented before Symington. Thus, if royal records in Spain may be trusted, a certain Blasco de Garay exhibited a steam vessel, in 1543, at Barcelona. He placed a large cauldron of boiling water in the ship, and a wheel on each side. Certain opinions concerning it were favourable, and Blasco was rewarded; but the invention was kept secret, and appears to have died.

Then, in 1655, the Marquis of Worcester is said to have invented something like navigation by steam. Later on, Jonathan Hulls took out a patent for a paddle steam vessel in 1736; and among others, in England, France, and America, the Marquis de Jouffroy made a steamer which was tried at Lyons, in 1783. Then, in 1787, Patrick Miller is said to have patented paddle wheels in Britain.

Miller was a retired gentleman at Dalswinton, in Dumfriesshire, who took much interest in mechanical affairs. He experimented with paddle wheels, and he also endeavoured to improve naval building. At first the wheels appear to have been turned by men, and there came a day when a double boat of Miller’s, worked by a couple of wheels with two men to turn each wheel, sailed with a Custom House boat,

and the need of more efficient motive power to revolve the wheels became very marked. Then the idea of steam navigation was born, or re-born.

There was a gentleman named Taylor, living with Miller, as tutor to his sons, and he often took part in the experiments with the boats. It is said that Taylor suggested the use of steam to propel the vessel, and that Miller doubted its practicability. However, he decided, at length, to try it, and in those summer days of 1787 the subject was much talked of at Dalswinton. Taylor mentioned the matter to Symington, who, it seems, was a friend of his, but it is not quite clear whether he had himself thought of this use of steam. However, in October, 1788, the experiment was tried on Dalswinton lake.

A boy was there who afterwards became Lord Brougham, and Robert Burns was also there; and, no doubt, the experiment was watched with much interest.

It appears to have been successful, and next year a bigger boat was tried on the Forth and Clyde Canal, again with some success. But whether Mr. Miller thought he had now spent enough money on these experiments—and Carlyle says Miller “spent his life and his estate on that adventure, and died _quasi_-bankrupt and broken-hearted”—or whether he was satisfied with the results attained, he abandoned all further effort. Possibly he did not see any opportunity of utilising the invention further. At all events, the development of the steamboat made practically no progress until Symington commenced his experiments under Lord Dundas.

Russell is of opinion that the invention of steam navigation was the joint production of these three men. “The creation of the steamship,” says he, “appears to have been an achievement too gigantic for any single man. It was produced by one of those happy combinations in which individuals are but tools, working out each his part in a great system, of the whole of which no single one may have comprehended all the workings.”

[Illustration: ROBERT FULTON.]

To these three, however, must be added Henry Bell, in Britain, and Robert Fulton, in America. They carried the great enterprise further on, to something like assured success.

Miller’s boats had two hulls, and the paddle wheels revolved between. Symington placed his wheel astern. Bell placed his paddles on either side.

“Ah, she will work!” we can imagine the spectators saying, as they watched that strange craft, the _Charlotte Dundas_, with her double rudder, tugging along her barges.

“Ay, she will work, but the canal folk won’t let her; they think the wash from the wheels will wear away the bank!”

“Then I will take the idea where it won’t be so hindered,” said another. “We are not afraid of our river banks in America.”

That man, whom we imagine said this, and who appears, without doubt, to have inspected the _Charlotte Dundas_, was Robert Fulton, who, with his companion,

Livingstone, claim to have invented steamboats in the United States.

This, then, in brief, seems to be the story. While bearing in mind the efforts of others, yet it would seem that Miller, Taylor, and Symington invented steam navigation, utilising improvements of Watt on the steam engine; but Fulton, in America, and Bell, in Britain, seeing something of these experiments, developed them to assured success.

What were Fulton's adventures?

CHAPTER II.

TO THE NARROW SEAS.

"I should not like to risk my money in the thing."

"Nor I, she will never pay."

"I reckon she will burst up before the day is over."

"Well, she is about to start now." A few minutes more, and the smiles on the faces of the speakers changed to expressions of astonishment. The boat was actually "walking the waters like a thing of life," and gathering speed as she drew away from the pier.

"Why, stranger, this thing's going to succeed."

"It does look so."

Still the speakers gazed, and still the vessel continued to glide along. And shouts and applause burst from the thronging crowd around. The "thing" was succeeding indeed.

They were watching the trial trip of the first practically successful steamboat in America, the _Clermont_. Fulton had been successful, and together with his companion, Livingstone—after whose residence the vessel was named—had launched a satisfactory steamer in America, five years before the _Comet_ appeared in Britain. Yet the _Clermont's_ engines were made in Britain by Boulton & Watt, and men from their works helped in mounting the machinery.

Colden, Fulton's biographer, describing this trial trip, says:—

"The minds of the most incredulous were changed in a few minutes—before the boat had made the progress of a quarter of a mile the greatest unbeliever must have been converted. The man who, while he looked on the expensive machine, thanked his stars that he had more wisdom than to waste his money on such idle schemes, changed the expression of his features as the boat moved from the wharf and gained her speed; his complacent smile gradually stiffened into an expression of wonder; the jeers of the ignorant, who had neither sense nor feeling enough to repress their contemptuous ridicule and rude jokes, were silenced for the moment by a vulgar astonishment, which deprived them of the power of utterance, till the triumph of genius extorted from the incredulous multitude which crowded the shores shouts and

acclamations of congratulations and applause."

The scene of the vessel's exploit was the famous river Hudson, and she came to make several trips between New York and Albany as a passenger boat. She performed the journey from Albany to New York in thirty-two hours, and back in thirty hours; her average speed being five miles an hour. Steamers now perform the passage in about eight hours.

The boat caused great astonishment at the time. Colden says she was described by some who saw her but indistinctly at night as "a monster moving on the water, defying the winds and tide, and breathing flames and smoke." He states:—"She had the most terrific appearance from other vessels which were navigating the river when she was making her passage. The first steamboats, as others yet do, used dry pine-wood for fuel, which sends forth a column of ignited vapour, many feet above the flue, and whenever the fire is stirred a galaxy of sparks fly off, which, in the night, have an airy, brilliant, and beautiful appearance. This uncommon light first attracted the attention of the crews of other vessels. Notwithstanding the wind and tide were adverse to its approach, they saw, with astonishment, that it was rapidly coming towards them; and when it came so near that the noise of the machinery and the paddles was heard, the crews in some instances shrunk beneath their decks from the terrific sight; and others left their vessels to go on shore; while others, again, prostrated themselves and besought Providence to protect them from the approach of the horrible monster which was marching on the tides, and lighting its path by the fires which it vomited."

Compare this with the stately passenger boats of the end of the century, gliding along four or five times as fast, but with little noise and less smoke, and beaming forth brilliant electric light from every saloon window.

The _Clermont_ was 133 feet long, 18 feet wide, and 7 feet deep. The cylinder of her engine was 24 inches in diameter, and her piston had a stroke of four feet; her paddle wheels were at first too large, or at all events dipped too deeply in the water. When improved they appear to have been fifteen feet in diameter. Her engines were 18 horse-power, and the tonnage was but 160.

Fulton was busily engaged in constructing steam vessels until he died in 1815. One of his efforts was the building of a steam war vessel; and so greatly were his efforts esteemed that both Houses of the United States Legislature testified their respect for him by wearing mourning apparel on the occasion of his death.

His work was developed by Mr. R. L. Stevens, whose father, indeed, had a steamer ready, only a few weeks after the success of the _Clermont_. Mr. R. L. Stevens came to grasp the idea that the form of the hull of steamships could be much improved by giving them fine lines instead of full round bows. Stevens, it is said, was able to obtain a speed of thirteen miles an hour; and he also, it is stated, used a different form of engine from that adopted by Fulton.

The engines of those early steamboats were, as a rule, a sort of beam engine. The famous _Comet_ was engined in that manner. John Robertson, who actually set up the _Comet's_ engines, lived to place them subsequently in South Kensington Museum. A

beam, or lever, which worked on a pivot at its centre, was placed between the piston on one side, and the connecting rod—which was fastened to the crank—on the other. Thus, one end of the beam, or lever, was attached to the piston rod, and the other to the end of the connecting rod which drove the crank and the wheel.

A development apparently of this beam-engine arrangement was the side-lever engine—a form of which marine engineers were also fond. The side lever seems, in fact, to have been a sort of double beam engine. The cylinder was placed upright, and a cross-piece was fixed to the end of the piston rod. From either end of this cross-piece a rod was connected with a beam or lever on either side of the machinery below. These levers worked on pivots at their centres, and their other ends were joined by a cross-piece united by a rod to the crank-shaft above. The idea in the side-lever engines appears to have been to obtain equal strength on both sides for each paddle wheel. Marine engineers did not apparently at first grasp the idea of a direct-acting engine—that is, simply one connecting rod between the piston and the crank which pulled round the wheel; perhaps the sizes and arrangements of those early steamboats did not permit of this. But in the development of the locomotive, the direct-acting engine did not appear at once. In any case, even the first vessels of the celebrated Cunard Line were of the cumbrous side-lever type.

Now, when Fulton had made his _Clermont_ in 1807, and Bell had put his _Comet_ on the Clyde, some of the English speaking people on both sides of the Atlantic began, we say, to see that there was a future before the new invention. In 1809, the _Accommodation_ ploughed the waters of the great St. Lawrence, and two years later a steamer startled the dwellers on the mighty Mississippi. The _Elizabeth_ also followed the _Comet_ on the Clyde in 1813.

She was bigger than her predecessor, but only of thirty-three tons; she was fifty-eight feet long, and her engine of ten horse-power. She was built by the constructors of the _Comet_, Wood & Company, of Port-Glasgow, under the direction of Mr. Thompson, who had been connected with some of Bell’s experiments.

The next step was the introduction of steamers on the Thames. All things gravitate to London, steamboats among the rest. Passing by some experiments, in which the names of a Mr. Dawson and a Mr. Lawrence appear, we find that George Dodd brought a steamboat from the Clyde to the Thames by sea, using both sails and steam, about the year 1813 or 1814. It is said that Dawson had a steamer plying between London and Gravesend in 1813, and that Lawrence, of Bristol, after using a steamer on the Severn brought her through the canals to the Thames, but was obliged to take her back because of the antagonism of the watermen. It is said also that the _Marjorie_, built by William Denny, of Dumbarton, was brought to the Thames about 1815 in six days from Grangemouth, having been purchased by some London merchants.

However this may be, the name of George Dodd should take a high place, perhaps next to that of Bell, for the enterprise and effort he showed in seeking to establish steam vessels. His sphere was chiefly the Thames, though he appears to have been also animated with the idea of using them upon the sea. The vessel he brought round

from the Clyde was named first the _Glasgow_ and afterwards the _Thames_, and was of about seventy-five tons, with nine feet paddle-wheels, and some fourteen or sixteen horse-power. He had some rough weather in the Irish Sea, and an account of the voyage is given in his book on steamboats. This, presumably in 1813, was the first steamship voyage at sea, as distinguished from steamers' voyages on rivers.

Such great progress had the introduction of steamboats made in 1818, that according to Dodd there were in that year eighteen on the Clyde, two on the Tay, two at Dundee, two at Cork, two on the Tyne, two on the Trent, two on the Mersey, four on the Humber, three on the Yare, one on the Avon, the Severn, the Orwell, six on the Forth, and actually two intended to run from Dublin to Holyhead. There may have been more than these, but they seem at all events to be the chief. Apparently there were, or had been, several on the Thames. Two, the _London_ and the _Richmond_, according to Dodd's book, were plying between London and Twickenham, and had carried 10,000 persons in four months. No wonder the watermen were alarmed.

Other vessels also had appeared on the royal river. The _Majestic_ even had got as far as Margate, and had ventured across to Calais. The _Regent_ had been burned off Whitstable, and the _Caledonia_, which had actually two engines, had steamed across to Flushing. Dodd further designed a vessel which seems to have gone to Margate in about seven and a-half hours, speeding along at about ten or eleven miles an hour. No wonder that Bell could say—"I will venture to affirm that history does not afford an instance of such rapid improvement in commerce and civilisation as that which will be effected by steam vessels." The _Richmond_ was a little boat of 50 tons, and 17 indicated horse-power. She was engined by Messrs. Maudslay & Field, of London, and presumably was the first steamer engined on the Thames. She ran from London to Richmond. In the next year Messrs. Maudslay engined the _Regent_ of 112 tons and 42 indicated horse-power, and intended to ply between London and Margate; while, in 1817, this famous firm engined three vessels, including the _Quebec_ of 500 tons and 100 indicated horse-power, intended for Quebec and Montreal. Since then they have engined hundreds of vessels, including screw-propeller ironclads of 20,000 horse-power.

Dodd, alas, though he worked so hard for the establishment of the steamship, does not seem to have profited by his labour. Like some other ingenious men he unhappily fell into poverty.

The next in order of succession, who apparently became the most prominent and among the most useful in the story of the steamship, was David Napier. Russell avers that from 1818 to about 1830 he "effected more for the improvement of steam navigation than any other man." David Napier ran the _Rob Roy_, a steamer of 90 tons and 30 horse-power, fitted with his own engines, between Greenock and Belfast. It appears that at one of the worst seasons he sailed in a vessel plying between the two ports,—sometimes taking a week to cover the journey, afterwards made in nine hours by steam,—and eagerly watched the effect of the heaving waves on the ship as she was tossed by the storm. Then, assured that there was no overwhelming difficulty for steamers, he started the _Rob Roy_. He also experimented upon the best shape of hull, and, without apparently any communication with Stevens across the Atlantic,

came to adopt a wedge-shaped bow, instead of a rounded fore front as common in sailing ships.

In 1819 he put the _Talbot_ on the Channel between Dublin and Holyhead. She was built by Wood & Company, and was one of the most perfect vessels of the kind then constructed. She had two engines of 60 horse-power combined, and was 150 tons burthen. She was followed by the _Ivanhoe_, and in 1821 steam vessels were regularly used to carry the mails.

Gradually the length of vessels increased without the beam being proportionately widened. The builders of those early boats did not at first realise the practicability and usefulness of altering the form of vessels for steamers. David Napier altered the bow, and gradually the vessels were lengthened. The idea came gradually to be grasped that as a steamer was forced forward along the line of its keel, and not by a power exerted upon it from without and in various quarters, its form might advantageously be changed. Moreover, it would seem that the best form for steamers is also the best for fast sailers. Russell is of opinion “that the fastest schooners, cutters, smugglers, yachts, and slavers” approach more nearly to the form of the best steamers than any other class of sailing vessels. However this may be, the shape of a steamer as well as its machinery has much to do with its speed, and David Napier appears to have contributed largely to these results in Britain.

Steamers had now sped out from the rivers into the narrow seas around Great Britain. The next step would be into the wide and open ocean. Who would venture to take it?

CHAPTER III.

ON THE OPEN OCEAN.

Why should not the Great Western end at New York?

That was Brunel’s idea, and it had an immense effect on the establishment of transatlantic steamships.

Brunel was the engineer of the Great Western Railway, and he audaciously desired his line to end, not at Bristol or Penzance, but, conquering the sea, he wished to plant his foot in the Empire city itself.

Still he was not the first, nor the only one, in the field. To the _Savannah_ belongs the honour of being the first steamship to cross the Atlantic. Yet she was not altogether a steamship.

Mr. Scarborough, of Savannah—a port of the state of Georgia—purchased a sailing ship of about 300 tons and 100 feet long, launched her at New York in 1818, intending her to ply between the two places, and had her fitted with machinery.

Why he changed his mind and sent her to Europe, we cannot say. Apparently he could not trust to steam alone, for the paddle wheels were so constructed that they could be folded up on deck when not in use, and the shaft also was jointed for that purpose. Then in the following May she started forth for Liverpool—the precursor of a

mighty fleet of magnificent ships which have followed since.

She reached the Mersey in twenty-five days—vessels now perform the journey in about six. But she used steam on only eighteen days out of the twenty-five. Several times during the journey the paddle wheels were taken on deck, this operation occupying about half-an-hour. Possibly this was done when the wind was very favourable for sails, and so saved the fuel, which was pitch-pine.

Apparently Mr. Scarborough was not satisfied with the venture, for, after failing to sell the ship in Russia, whither she voyaged, she touched at different ports and returned home. The machinery was taken out, and she winged her way henceforth by sails alone.

England next did something of the same kind. The _Falcon_ steam yacht, a little vessel of 175 tons, voyaged to India in 1824, mostly, however, by the power of sails. In the next year the _Enterprize_, engined by Messrs. Maudslay & Field, made the passage by steam to Calcutta from London in the net time of 103 days—ten being used in stoppages, and the entire voyage thus occupying 113 days. She was a vessel of 500 tons, 122 feet keel, and 27 feet broad, while her engines were of 240 indicated power. Then the _Royal William_, hailing from Quebec, made the transatlantic passage in 1831, principally by steam, in twenty-six days. In 1835 Messrs. Willcox & Anderson began to run steamships to Peninsular ports—an undertaking which blossomed out afterwards into the celebrated Peninsular and Oriental Steamship Company.

Then in 1838 two steamships, the _Sirius_ and the _Great Western_, crossed the Atlantic, the latter in fourteen and a-half days. Brunel had had his wish, and in 1836 he had formed the Great Western Steamship Company, and the vessel of the same name had been commenced. Others also were in the field, notably Messrs. Laird of Birkenhead, and the British and American Steam Navigation Company was founded. The _Sirius_, which had been built on the Thames, was purchased by them and prepared for her voyage.

The prime mover in this matter is said to have been Mr. Macgregor Laird. He had witnessed the work of steamships in the Niger Expedition of 1832-33 both on sea and river, and from the time of his return he advocated the establishment of steamships between Great Britain and America.

The _Sirius_ left Cork on the 5th of April, and arrived at New York eighteen days afterwards. She carried seven passengers, and close at her heels followed Brunel’s _Great Western_, which had left Bristol three days later. The two ships were received with loud acclaim, a vast crowd of spectators beholding their arrival. The vessels proved beyond possibility of doubt that the transatlantic voyage by steamships was possible, and, at a stroke, the duration of the passage was reduced by almost one-half. It has since been reduced to less than a quarter.

The _Sirius_ made on an average about 161 miles a-day, or slightly less than seven miles an hour. She apparently, however, had been originally built for plying between London and Cork; while the _Great Western_, which had presumably been especially built for the transatlantic traffic, was both larger and more powerful. Her average

speed was about 208 miles a-day, that is between eight and nine miles an hour; while returning, the speed was a little better, averaging about 213 miles per day. The return voyage of the _Sirius_ was also better than her outward passage.

The engines of the _Great Western_ were side-lever, and were built by Messrs. Maudslay & Field, of London. The cylinders were 73½ inches diameter, and the pistons had a big stroke of seven feet. The wheels' diameter was no less than 28¾ feet, while the steam was generated in four boilers. Her tonnage was 1340—the largest Maudslay's had yet engined, with 750 indicated horse-power. She voyaged many times across the Atlantic, her fastest eastward passage being 12 days, 7½ hours. The variation in her coal consumption was very remarkable. Thus, on her first voyage 655 tons were burnt, but on her return journey she consumed 263 tons less. No doubt this was owing to the greater use she was able to make of the wind.

The proprietors of the two vessels soon began to build others. The owners of the _Great Western_ laid down the _Great Britain_, and the proprietors of the _Sirius_ began the _British Queen_. She had paddle wheels of 31 feet diameter, and her piston stroke was the same as the _Great Western_, 7 feet. Her engines were 500 horse-power, and her cylinders 77½ inches in diameter. She was 275 feet long, 40 feet wide, and 27 feet deep. From Portsmouth to New York she crossed in 14 days, 8 hours.

Satisfactory as these results were, the pecuniary returns unfortunately were not so favourable. The _Great Western_, it is said, continued running at a loss, but others were withdrawn. Something seemed wanting to make the venture a commercial success. What was it?

Meantime Willcox & Anderson's steamers plied with remarkable regularity to the Peninsula, and this regularity aroused some attention. The Government of the day applied to the proprietors to submit a scheme for carrying the mails. It seems that previously Willcox & Anderson had proposed this, but it had come to nothing. The end of the matter was, however, that the first mail contract was signed with them, the 22nd of August, 1837. To carry out their bargain, Captain Richard Bourne and Messrs. Willcox & Anderson founded the Peninsula Company, and three years later it was expanded to the Peninsular and Oriental Steam Navigation Company—popularly known as the P. & O.—and incorporated by Royal Charter. The mail service was the keystone of the enterprise.

The first steamer, built in 1829, was the _William Fawcett_, a small vessel of 206 gross tonnage, and but 60 horse-power. In 1842 the proprietors owned the _Hindostan_, of 2017 gross tonnage, and 520 horse-power. She was a paddle-wheel vessel, and opened the Indian Mail Service. The commencement of this service marks another stage in the history of steam navigation. About fifty years later the Company owned about half-a-hundred ships, two being of 8000 horse-power and 7000 tonnage.

Some two years after the _Hindostan_ first steamed to India, Brunel's _Great Britain_ was finished. She was a very remarkable vessel, and the wonder of her time. In the first place, she was built of iron, and, secondly, she was propelled by a screw, though at first it was intended that she should have paddle-wheels, and the engines

for these wheels had been partly made.

Barges and light vessels had been built of iron since about 1790, or earlier, and the Lairds of Birkenhead, among others, had built an iron vessel about 1829. It is said that the _Aglaia_ was the first iron steamer built on the Clyde in 1832. As for the screw-propeller, John Ericsson was successful with the _Francis B. Ogden_ in 1836, and three years later Sir Francis Pettit Smith clearly showed, in the vessel appropriately called the _Archimedes_, the value and the feasibility of the new system.

Brunel, therefore, ever open to improvements, combined these two alterations in the _Great Britain_. It was in 1839, probably after Sir Pettit Smith's success, that the change was made as regards the screw for this vessel, though the paddle-wheel engines had been begun. The superiority of the screw-propeller over the paddle-wheels are said to be these:—the engines occupy less room, and are lighter—two very important considerations. Then there is greater wear and tear on paddle-wheels, and consequently the screw vessels are less expensive. But most important of all, the screw being deep in the water, the vessel is much more suitable for ocean traffic. In the heaving billows of the sea one wheel may be buried deep on one side of the ship, and the other whirling round high in the air, and not propelling the vessel; whereas the screw, being always immersed, except possibly in severe pitching, is more constantly efficient for the whole of the vessel.

Nevertheless, paddle-boats have their advantages. They need less water to work in, are started more easily, and stopped sooner. Further, it is said they are less liable to cause sea-sickness, as they do not roll so much. In a word, the difference seems to be this: paddle vessels are better suited as passenger boats on the shallower waters; screw vessels for deep sea and long distance voyages, though whether the adoption of twin-screws,—which it appears need not be immersed so deeply in the water as one screw,—will bring screw vessels into use on shallower waters remains to be seen.

But when the _Great Britain_ was being built the greater efficiency of the screw-propeller for ocean voyages was not widely understood. She was a fine vessel, over 320 feet long, 51 feet wide, and 32½ feet deep. Her screw was successful; but on her fourth voyage to New York she became stranded in Dundrum Bay, and lay aground for nearly a year.

Incidentally, however, this catastrophe seems to have given great impetus to iron shipbuilding; for after being floated, she was discovered to have suffered but comparatively slight damage. She was seen in dock by many persons interested in shipping, and they became impressed with the practicability and usefulness of iron for shipbuilding.

Unfortunate _Great Britain_! She passed through many vicissitudes. Her owners got into difficulties, and after some alterations, she ran to Australia, and at length she wheezed her way to the Falkland Islands, where, it is said, she served as a hulk—a sorry end to a successful beginning.

The engines of the early screw vessels appear to have very much resembled those for paddle-wheels ships. Thus the _Rattler_, engined by Messrs. Maudslay for the

Admiralty about the year 1841, had upright cylinders, with a crank-shaft overhead and wheels to give speed to the screw.

In the meantime, however, the commercial difficulty of transatlantic steam traffic was being solved. The something lacking had been supplied. What was it?

CHAPTER IV.

THE OCEAN RACE.

“This is the very opportunity I have been wanting!”

The speaker was looking at a paper setting forth that the British Government were open to consider contracts for the carrying of the letters by steamships between Great Britain and America. Encouraged, no doubt, by the success attending the conveyance of the mails by similar means to the Peninsula, the Government were now going farther afield.

The practicability of ocean steam traffic had been amply demonstrated; but some of those early steamships did not “pay,” and to that test, after all, such undertakings must come. Now, the man into whose hands the circular had fallen was of great intelligence and remarkable energy. He was a merchant and owner of ships, and agent for the East India Company at Halifax, Nova Scotia. His name has since become known the wide world over. It was Samuel Cunard.

Apparently he had cherished the idea of establishing transatlantic steam traffic for some years—since 1830 it is said—and now, here was the opportunity. The British Government would, of course, give a handsome sum for carrying the mails, and that sum would form a backbone to the enterprise.

Over came Cunard to London in 1838. Mr. Melvill, the secretary of the East India Company, gave him a letter of introduction to Mr. Robert Napier, the eminent engineer at Glasgow. Thither then went the indomitable merchant, and was heartily welcomed. Napier knew Mr. George Burns, who was partner with Mr. David MacIver in a coasting trade, and the upshot of the matter was that capital of considerably over a quarter of a million (£270,000) was subscribed through Mr. Burns’s influence.

The first great step thus taken, Mr. Cunard made a good offer to the Government, and although another offer was made by the owners of the _Great Western_, Cunard got the contract, the tender being regarded as much more favourable. The subsidy was eventually £81,000 per annum. The contract was for seven years, and was signed by the three gentlemen mentioned—Cunard, Burns, and MacIver.

These three divided the labour. Cunard ruled at London, MacIver at Liverpool, and Burns at Glasgow. Napier was to engine the new vessels. It was decided that their names were all to end in “ia,” and nearly every one of the now historic fleet has rejoiced in a title of that ending. There is a sailor’s superstition that it is unlucky if the vessels of a fleet are not named with some uniformity; but we doubt if the superstition influenced the Cunard Company. In any case, they broke another superstition by starting their first ship on a Friday! She was a mail ship, and she had

to go. The Cunard Company meant business.

But about their fleet. Their first order was for four vessels, all of about the same size and power. The _Britannia_ was the first, and her sisters were the _Caledonia_, the _Columbia_, and the _Acadia_. They were paddle steamers, the value of the screw not having then been clearly and widely demonstrated, all of them about 207 feet long, 35⅓ feet broad, 22½ feet deep, and 1154 tons burthen. The engines—side-lever, of course, in those days—were of 740 horse-power. The boilers had return-flues, and were heated by a dozen furnaces.

They would look now quite out of fashion, like a lady's dress of a past age. They appeared something like sailing ships, with the straight funnels added.

The _Britannia_ began the service by starting from Liverpool on the 4th of July, 1840, and, attaining a speed of about 8½ knots per hour, she made the passage to Halifax in 12 days, 10 hours, and returned in 10 days. Her average consumption of fuel was about thirty-eight tons daily.

The Bostonians gave the _Britannia_ quite an ovation. A grand banquet, followed by speeches, celebrated the great occasion. But they gave even more practical appreciation of their favour subsequently, for when, in the winter season, the vessel became ice-bound in the harbour, they cut a seven-mile passage for her through the ice, at their own cost.

The Cunarders were successful, and the conveyance of the mails by steamship became quite established. The white-winged clipper ships fought hard against the Cunarders, but they had to yield. Three years later the Company put another vessel on the route—the _Hibernia_—and in 1845 the _Cambria_. These were of greater size and developed a little better speed than their forerunners. It has always been the policy of the owners to improve their ships as they went on building, and even thus early that policy ruled.

The establishment of the Cunard Company marks a most important step in ocean steam navigation. Further, in the same year, 1840, in which the Cunarders began to run, the Pacific Steam Navigation Company was established. Ten years later saw the foundation of the Collins and the Inman Lines. The Collins, an American Line, boasted that they would run "the Cunarders off the Atlantic." They were very fine vessels, and they were the first fleet to fully adopt the upright stem and discard the bowsprit. But the Cunarders were ready for the fierce competition. They had actually put on six new vessels, and their new postal contract of 1847 had stipulated for a weekly, instead of a fortnightly service; while the subsidy was much increased. It was to be £173,340 annually instead of £81,000.

[Illustration: THE ICE-BOUND "BRITANNIA" AT BOSTON. _By permission of The Cunard Steamship Co._]

The echoes of that fierce struggle between the Cunarders' and the Collins' boats have now died away, or have been quite lost in the other clamorous cries of that wonder of the world, the development of the transatlantic steamship traffic; but apparently partisanship ran very high. The Collins' seem to have been slightly the

faster vessels, coming from America in 9 days 17 hours, but occupying nearly two more days to return. Alas, disaster overtook them. The _Arctic_ perished by collision; the _Pacific_ was lost at sea, and no one knows the story of her death, for she was never heard of more. Bad management, and extravagance surged over the remaining vessels, and the fine ships went as old iron!

But the Inman line had also begun to run, about 1850. These ships, like the _Great Britain_, were built of iron and propelled by a screw. The first was the _City of Glasgow_, and several famous “Cities” followed; though years afterwards the Inman line became the “American,” and the appellation “City” was dropped, the ships being simply known as _Paris_, _New York_, _Berlin_, etc. The Inman line had the distinction of being the first, apart from the _Great Britain_, to use iron screw steamers regularly on the Atlantic. Other lines soon followed, the Anchor, the Allan, and the Guion, while the Cunarders, not to be beaten, came along in due course with iron and screw steamers.

But great changes were at hand. To mark these changes let us look at what may be called the culminating ship of the old type of steamers—the _Great Eastern_.

This historical vessel was the largest ever built. She was 680 feet long, by 83 feet broad, and her hull was 60 feet high, 70 feet including bulwarks. But the steam pressure of her engines was only from 15 to 25 lbs. She was fitted with both screw-propeller and paddle wheels. Her screw-propeller engines were of 4000 indicated horse-power, and paddle of 2600, but they could together work up to 11,000 horse-power.

Commenced at Millwall early in 1854, she was not launched until near upon four years later. The launching itself was a difficult and expensive business, costing £60,000, and only effected after various attempts extending over nearly three months. The total cost of the vessel has been estimated at £732,000.

[Illustration:

ISAMBARD KINGDOM BRUNEL. _By permission of Messrs. Graves & Co._]

It will be seen at once that so large an outlay required an immense business to yield a satisfactory return, and indeed, financial difficulties hampered her success almost from the very commencement, even before she was launched.

She was planned, in 1852, by the great engineer, I. K. Brunel, and by Scott Russell. In the life of Brunel by his son, it is stated:—“It was, no doubt, his connection with the Australian Mail Company that led Mr. Brunel to work out into practical shape the idea of a great ship for the Indian or Australian service.”

The Eastern Steam Navigation Company desired a vessel to trade to Australia and back, large enough to carry a sufficiency of coal for the outward and homeward journey, and yet to have space for a goodly number of passengers and a bulky amount of cargo.

That was the idea, and we perhaps can hardly realise what a difficulty this question of coal carrying capacity was in those days, before the problem had been solved by

high-pressure steam boilers, triple expansion engines, improved condensation, and quick passages. Even so great a philosopher as Dr. Lardner could not believe in 1835 that a steamship could voyage from Liverpool to New York without stopping—we presume for fresh fuel.

The _Great Eastern_, therefore, was planned to carry 15,000 tons of coal; whereas now the large Atlantic liner _Paris_ needs only 2700 tons for her Atlantic trip. The difference is most striking, for the _Paris_ is one of the largest steamships afloat, but her working steam pressure is 150 lbs. instead of the 15 or 25 lbs. of the _Great Eastern_.

This immense vessel was also planned to carry some 5000 persons, or about 500 less if any large number were to require state-rooms, and finally she was to convey 5000 tons of cargo. The idea of water-tight compartments was anticipated in her case, even to the extent of longitudinal ones, and she had half-a-dozen masts of which five were of iron.

When at length she was launched, the directors' minds misgave them as to an Australian trip, and they determined to cross the Atlantic instead, for a trial voyage. She started on the 8th of September, 1859, but alas! when off Hastings some steam pipes burst. Several persons were killed and wounded, and the voyage ended at Portland.

Next year she tried again and crossed in eleven days, after which she made several voyages with success—on one occasion conveying soldiers to Canada. Unfortunately for the owners, however, she did not pay.

Then in 1865 she began to be engaged in submarine telegraph work, by which she will most likely be best remembered, and two years later she was chartered to convey passengers from America to Havre for the French Exhibition, but this scheme failed.

Then for some years from 1869 she was successfully engaged in cable-laying, in the Red Sea, the Atlantic, and the Mediterranean, etc., after which she came down to be a coal hulk in 1884, stationed at Gibraltar.

At length she was sold for £26,200 at London, by auction, and was on view in the Thames, and also in the Mersey. At this latter river her huge sides were used as an advertising "board" for a Liverpool business house. Again in November, 1888, she was sold by auction, this time for breaking up, and it is said that the total proceeds of the sale which lasted five days was £58,000, more than double what she had previously brought!

"A ship before her time," says some one, thinking of the huge vessels of the last decade of the nineteenth century. That is true, but the immense space required for coal, and her low-pressure engines, had also something to do with her comparative failure. The problem which the _Great Eastern_ failed to solve has been met in other ways—viz., by the use of high-pressure steam and compound, triple-expansion and even quadruple-expansion engines. That is, the steam, working at 150 or 160 lbs. pressure, instead of the 25 lbs. of the _Great Eastern_, is passed through two, three, and even four cylinders respectively, and the economy in coal consumption is

astounding. Thus the use of triple expansion engines has brought the saving in coal down from 4 lbs. per indicated horse-power to less than 1½ lbs.

There have been many other improvements also, such as the use of steel instead of iron, the parts being thus stronger and yet lighter; the circular tubular boiler enabling high-pressure steam to be economically produced and maintained; the use of surface condensers, by which the exhaust steam is quickly reduced to water and returned to a “hot well” ready for the boilers, to be speedily again raised to high pressure steam; and a forced draught by which the furnaces are made to roar furiously and heat the water in the boilers speedily.

[Illustration: THE “GREAT EASTERN.”]

But these things were not all attained in a day. The introduction of the compound marine engines in 1854-56 by John Elder, marks the first great step of the new departure. In 1856 he engined vessels for the Pacific Steam Navigation Company, on the compound principle, which proved very satisfactory.

Again in 1870, the appearance of the White Star liner _Oceanic_, marked a new development. Her yacht-like shape, great length, and general symmetry of form commenced a marked change in Atlantic liners.

It was in 1867 that Mr. T. H. Ismay bought the interest of the managing owner of the White Star line—a set of sailing clippers, dating from the rush to the Australian gold diggings—and began to introduce iron vessels instead of wooden clipper ships. In 1869 he established the Oceanic Steam Navigation Company—popularly known as the White Star—and was later on joined by Mr. William Imrie. The Company was started with so much wisdom and boldness that the £1000 shares were privately taken up at once. The order for the new steamers was given to Harland & Wolff, of Belfast, because, it is said, an influential share-holder had had satisfactory dealings with them before.

The _Oceanic_ was of 3600 tons burthen, and with engines of 3000 horse-power. The accommodation for first-class passengers was placed amidships, where the motion of the vessel is said to be felt the least, and altogether she embodied improvements which made her the type of many of the Atlantic passenger ships since. The earlier White Stars were fitted with compound engines, and reduced the passage to about 8½ days.

But when the White Stars _Germanic_ and _Britannic_ appeared in 1877, then a marked advance indeed was made in the Atlantic record. The _Britannic_ astonished the world by speeding from Queenstown to New York in 7 days, 10 hours, and 50 minutes, and since then she has beaten her own record. Her sister ship _Germanic_ also did as well, and the fierce race for the blue ribbon of the Atlantic may be said to have begun.

It was even prophesied that the time across the water might be reduced to six days. How has that been fulfilled?

CHAPTER V.

“The record’s broken again, Jemmy! The White Star has come home a couple of hours earlier!”

“She has, has she? Well, it will be the Cunard’s turn next week. It’s wonderful what they get out of the Cunard’s engines.”

“They do; but I’m thinking the American’s _New York_ will be doin’ the fastest bit.”

“Well, well, it may be. They’re all main powerful vessels. Do you mind when the Guion’s _Alaska_ came home in 6 days, 18 hours, 37 minutes?”

“I do, and about ten years later, I suppose, some ships were doing it in about a day less time!”

“Ay, ay, and I see they’re goin’ ahead down south too.”

“Yes, there’s fast steaming all over the world, Jemmy!”

“I told you what would happen when the compound engine came into use. I said, ‘Mark my words, now they’ve got the compound engine, they will go ahead’—and they have.”

Jemmy’s prediction has been amply verified, for almost every year since the compound engine came largely into use, has witnessed a greater speed in ocean steamers.

And the speed has not been obtained at sacrifice of comfort. On the contrary, an ocean passenger steamer belonging to any of the great passenger lines is something like a floating palace.

After the _Britannic_ and _Germanic_ appeared, line after line put forth fine vessels; and in 1889 was launched the White Star steamer _Teutonic_, which for some time held the proud position of the fastest ship on the Atlantic. She had crossed in 5 days, 16 hours, 31 minutes. The average of several trips, both for herself and her sister _Majestic_, was 5 days, 18 hours, 6 minutes. And they were run very close by the American liners, _Paris_ and _New York_. These four vessels were among the first propelled by twin-screws. Engineers began to see that it was better to use great power in two shafts and two propellers than in one.

In July, 1892, the fine Inman (now called American) liner _Paris_ crossed the Atlantic in 5 days, 15 hours, and 58 minutes, and in October of the same year the same vessel steamed from Liverpool, touching as usual at Queenstown, in 6 days, 2 hours, and 24 minutes—including the time at the Irish port. This was then the quickest time on record for the entire journey. From Queenstown to Sandy Hook the time was 5 days, 14 hours, and 24 minutes, a gain of 1 hour and 34 minutes on her voyage in the previous July. Her best day’s run was 530 knots.

The contest, therefore, between the two White Stars and the two Inmans has been very close, the record time resting now with the one and then with the other.

But the Cunard Company, not to be beaten, put on the _Campania_ in 1893, and in April of that year she made the fastest maiden trip then on record, one day indeed compassing 545 knots in the 24 hours.

The _Campania_ is 625 feet long by 65¼ feet broad, and 43 feet deep from the upper deck. Her gross tonnage is 12,950. She is fitted with a cellular double bottom extending fore and aft, and also with sixteen bulkheads, so arranged that the vessel would float even if two, or in some cases three, compartments were open to the ocean.

She is a twin-screw vessel, fitted with two sets of very powerful triple-expansion engines. They are seated in two separate engine-rooms with a dividing bulkhead and water-tight doors.

Each set of engines has five inverted cylinders—viz., two high-pressure, one intermediate, and two low-pressure—all arranged to work on three cranks set at an angle of 120 degrees to each other. Her indicated horse-power is 30,000. The boiler-rooms are doubly cased, the space between being fitted with nonconducting material for sound and heat.

[Illustration: HIGH AND LOW PRESSURE CYLINDERS OF THE “CAMPANIA’S” ENGINES.]

In this huge vessel four decks rise tier above tier, beside erections on the upper deck, known as promenade and shade decks. These four principal decks are the orlop, the lowest of all, used for cargo, stores, and machinery; the lower, the main, and the upper decks, the last three being devoted entirely to passengers.

Imagine yourself on the upper deck. Before you stretches the long vista of its length, like some far-reaching walk ashore; a circuit of the vessel four times makes a mile. Above rises the shade deck with the navigating apparatus, and surrounded by the twenty lifeboats of the vessel; above again is the captain’s bridge, where are placed the telegraph and wheel house, while higher still is perched the crow’s nest or look-out box, on the foremast, and about 100 feet from the water-level. Give a glance, too, at the huge funnels, 120 feet high, and so large that when in the builder’s yard a coach full of passengers was driven with four horses through one of them.

Descending then, the grand staircase, which is sufficiently wide for six persons to walk down abreast, and admiring the polished panelling, the rich Japanese paper, and the lounges on the landings, we enter the superb dining-saloon 100 feet long by 62 feet broad. Four huge tables run almost along its length, with smaller tables in the corners, while the wood-carving, carpeting, gold decorated roof, costly mirrors, and upholstering in rich red velvet are of the most sumptuous description.

From this magnificent hall you can wander on through other apartments of great splendour, drawing-room, library, smoking, music room, bath-rooms, and numbers of state-rooms. There are single berth, double berth, and three and four berth cabins—the old wooden benches for beds, however, being replaced by iron bed-steads throughout the ship. The electric light glows everywhere, being distributed by some fifty miles of wire.

[Illustration:

THE “CAMPANIA.” _By permission of The Cunard Steamship Co._]

The second-class accommodation differs but in degree from the magnificence of the saloon, while the steerage passengers are berthed on the lower deck, but have the privilege of walking on the upper deck. An additional idea of the size of the ship may be gained when we learn that the crew consists of over 420 persons—viz., 190 engineers, 179 stewards, and 54 sailing hands, while the vessel’s full complement of passengers brings up the total number of persons aboard to 1600 souls—quite a floating town indeed.

About five years after the birth of the _Teutonic_ the newspapers recorded, in May, 1894, that the _Lucania_, sister ship to the _Campania_, and one of the newest Cunarders, had performed the journey across the Atlantic in 5 days, 13 hours, and 28 minutes. Her average speed was 22¼ knots, or 25·7 land miles per hour, marking one of the quickest runs then ever recorded; and about the same time came the news that the P. & O. steamer _Himalaya_ had completed a mail transit from Bombay of 12½ days, and as her voyage to Bombay had been just over 13 days—the best outward passage—she had completed a round mail transit to Bombay and back, excluding stoppages, of 25½ days.

A little later, in the same year, the torpedo-boat destroyer, _Hornet_, built by Messrs. Yarrow & Co., of Poplar, for the British Navy, achieved, it is said, about 27 knots; that is, roughly speaking, near to 29 or 30 miles an hour, which speed proclaimed her to be then one of the fastest steamships in the world. She was fitted with the Yarrow water-tube boilers, which are both light and strong, while the consumption of coal was said to be remarkably small. She has two sets of triple-expansion inverted engines.

Again, a short time later, Messrs. Thorneycroft, of Chiswick, obtained similar results with the _Daring_, another boat of the same kind built for the British Government, and fitted with the Thorneycroft improved water-tube boilers. These, it is claimed, will raise steam from cold water in fifteen minutes. She passed the measured mile on the Maplin at the high speed of 29¼ miles an hour.

In the same summer a Company put on a fine steamer for service on the Thames and the English Channel, called _La Marguerite_, which developed, it is said, a speed of 25 miles an hour, which would make her one of the fastest passenger vessels then afloat.

Another Company has also a noteworthy vessel running on the Estuary of the Thames—viz., the _London Belle_, plying from London Bridge to Clacton-on-Sea. She is a triple-expansion paddle boat, and the first river steamer fitted with three crank triple-expansion paddle engines. She was built by Denny of Dumbarton, and can develop a speed of 19½ knots—_i.e._, twenty-three statute miles per hour, and is worked with great economy of coal consumption.

An example of a quadruple-expansion engine steamer may be found in the _Tantallon Castle_, one of the newest vessels for voyaging to South Africa. She is 456

feet long, over 50 broad, with a gross tonnage of 5636. She is fitted with quadruple-expansion engines of 7500 horse-power, and the stoke holes are well ventilated by large fans speeding round with great swiftness.

Improvements in steamship building had gone steadily on; and it is safe to say that a pound of coal, after the compounding principle came fully into use, did four or five times the work it accomplished before high pressure engines were fully utilised.

Let us enter the engine-room of a big liner, and see for ourselves. It is a triumph of engineering. Still, at first, you cannot understand anything of the complicated mass of machinery. Then you notice three large cylinders—for these are triple-expansion engines—with pistons shooting in and out downwards, and attached by connecting rods to the cranks of the propeller shaft below. The cranks are bent at different angles so that they can never all be in the same position at once. There is a maze of machinery and shining rods, bewildering to the uninitiated eye. But you gradually notice how absolutely regular every part is in its action, and how beautifully one part fits with another.

Then go before the furnace; you find yourself in front of a huge structure, at the bottom of which is the long fire box; above rises the heat box communicating with tubes over the furnaces, with the water circulating between. The water, indeed, is beneath the furnace, about parts of the heat box, between and above the tubes. The object is, of course, to obtain as great heating surface as possible. The tubes communicate with the funnel at their other end. Boilers are made of a “mild” steel which has, it is said, a most remarkable tenacity of 28 tons to the square inch. Consequently they are able to bear great pressure of steam.

[Illustration: STOKE HOLE.]

Hot distilled water is admitted to the boiler from the surface condenser. This is a “box,” riddled with tubes, through which cold sea water is pumped. The waste steam, having done its work in the cylinders, is passed into this “box,” is condensed by touching the chilly tubes of sea water, and can be run off or pumped to a hot cistern, whence it is used to feed the boiler and be turned once more to steam. About 4000 tons of water an hour pass through the surface condensers of a large liner when she is at full work.

The largest steamers require over 150 men to work the furnaces and machinery, and the attention given is hard and unremitting. In some of the fast Atlantic greyhounds the strain is terribly severe, especially when the sea is beginning to run high. The rollers may be but 20 feet, yet these are quite high enough even for a splendid ocean racer to contend with and yet maintain her speed.

Now her bows are pointing sky high, and her stern is deeply submerged; now she takes a header plump into the trough of the sea, and the engines race round; the propeller is suddenly raised out of water. But blow high, or blow low, on she goes, and the engineers are always busy. The furnaces roar with ceaseless rage. For days and nights the fires are kept at glowing heat. A forced blast maintains the draught; the steam condensed back into warm water is supplied to the boilers; half-naked men work hour after hour to rake the fires, clean them, pile on the fuel, and keep the most

powerful head of steam the boilers can stand.

When the furnace doors are opened tongues of flame leap forth, and the heat is enough to make a man sick. But with head turned away, the stoker stirs up the fire with his huge “slice” or fire rake, and cleans out the clinker clogging the bars.

Then on go the coals! One layer, shot in from the shovel with unerring precision and skilful experience, right at the back; then another just in front of the first, and so on till the long furnace is filled. Bang! the furnace door clangs, and the man reels away, sick and exhausted, with tingling eyes and heaving chest. Then coal has to be brought from the bunkers to the furnaces, tons of it per day, and if the ship rolls too much for the barrows to be used, the fuel must be carried in baskets.

There is an engineer in charge of each stoke hole, and two on the platform in each engine room; as a rule, the staff are on duty in turns—four hours out of every twelve. But if the weather be bad they may have harder times.

No matter how hot the machinery becomes, the engineers must not reduce speed, except it be to prevent disaster. Oil is swabbed on in bucketfuls, so to speak, but at every thrust the polished steel may gleam dry and smoking. Then on goes the water, as if there actually was a conflagration, and meantime a mixture of oil and sulphur is dabbed on. The water flies off in steam, so hot are the bearings, so terrific the friction of the incessant speed; and at last, down comes the reluctant order, wrung out of the chief like gold from a miser—“Slow her down.”

It is done—dampers are clapped on furnaces, steam pressure dropped a little, and engines reduced to half speed; the three great cranks of the high, intermediate, and low pressure cylinders move round easily, and the tremendous noise gradually sinks to a murmur, compared with the previous rush and roar. The machinery cools. But when quite safe, on is piled the speed once more, and again the cranks fly round, and the mighty engines work their hardest to drive the mammoth ship through the surging green rollers.

So superbly are these marine engines built, and so excellently are they maintained, being continually overhauled, so as to be kept in the pink of perfection, that, as years go on, they seem to “warm to their work” and do even better than at first.

On the completion of the 200th round voyage of the celebrated “White Stars,” _Germanic_ and _Britannic_, about January, 1894, they seemed steaming as regularly and as fast, or faster than ever. Thus, on the 198th outward trip of the _Germanic_, in September, 1893, she made the fastest westward passage, but one, she had ever accomplished. During their lives, it was said these vessels had maintained remarkable uniformity in speed, and each vessel had steamed 200 times 6200 nautical miles, that is nearly a million and a-half statute miles, with the original engines and boilers—a performance, in all probability, without parallel in the world.

Those people who care for figures may be interested in knowing that the _Britannic_ had been 91,741 hours under steam, and 85,812 hours actually under weigh. Her engines had made 280 million revolutions, and maintained an average speed of 15 knots, or 17¼ statute miles an hour, while she had burnt 406,000 tons of

coal. During their nineteen years of life the two vessels had carried 100,000 saloon, and over 260,000 steerage passengers, in safety and in comfort.

This is a record of which all concerned, builders, owners, and working staff, may well be proud. It augurs first-class, honest work, and superb engineering skill. Since the construction of these ships, however, vessels surpassing them in speed have, of course, been built, among which may be mentioned the same line's _Teutonic_ and _Majestic_.

The well-known Cunarders, _Umbria_ and _Etruria_, have also done some very fine work, indicating great excellence of construction. Thus, on her eighty-second voyage, the _Umbria_ steamed from Queenstown to Sandy Hook in 5 days, 22 hours; or, allowing for detention through fog, 5 days, 18½ hours, which is within three or four hours of the White Stars' and American's time.

The story of the British warship _Calliope_, at Samoa, will also show how marvellously well ships' engines can be built. Some difficulties had arisen between the United States and Germany as to Samoa, and several warships had gathered there. Some weeks of bad weather had occurred, and then, on the 15th of March, 1889, the wind began to blow with tremendous force. Down came the top masts from the warships—taken down as a precaution; steam was raised in the boilers in case anchors should not hold, and spars were made secure. But no man among the sailors expected such a hurricane as ensued.

Rain fell at midnight, and the wind increased. Huge waves rolled in from the South Pacific, and the vessels tugged madly at their anchor chains and pitched fearfully up and down, like corks. Then the _Eber_, one of the German ships, began to drag her anchors; and the _Vandalia_, one of the Americans, followed suit. But by their steam power they kept off a dangerous reef, and also prevented themselves from colliding with their neighbours.

Still higher and higher blew the hurricane, and the rain fell with tropic severity. Three hours after midnight the situation had become terrible. Almost every vessel was dragging her anchors, and the danger of collision was constant.

The scene of the occurrence was a small bay before Apia, the capital of Samoa. But there is a coral reef extending in front of the bay for about two miles, and in the centre of the reef an opening about a quarter of a mile wide. The ships, therefore, were shut up in a comparatively small space, from which the way of escape was this gateway through the reef. The tide rushed in with great rapidity, swamping the land a hundred feet or so above high-water mark.

As morning dawned and wore on to-day, the _Eber_ collided with the _Nipsic_ and then with the _Olga_, and, finally, was dashed by the huge waves, like a toy, upon the reef, and rolled over into deep water. Only five men struggled to shore and were saved. Other sad disasters occurred; and then, shortly before noon, the _Vandalia_ and the _Calliope_ were tossed perilously near together, and also toward the dangerous reef. In endeavouring to steam away, the _Vandalia_ collided with the _Calliope_, and was much damaged. Then, with splendid courage, Captain Kane determined to steam right away to sea—to remain would but risk another collision, or

a wreck on the reef. Sea-room he must have at any cost!

“Lift all anchors!” was the thrilling order, and then—“Full speed ahead!” Round swung the vessel’s head to the wind, and though the powerful engines were working “all they knew” to force the ship along, the steamer stood still, as if aghast at being asked to break through these tremendous waves.

But she stood for a moment only. The superb engines began to tell; the quickly-whirling screw churned up the heavy water at the stern, and slowly the good ship made headway through the huge billows. They crashed over her stern and poured over her decks, as if in anger at her defiance. But on went the coal to her furnaces, and the thick smoke reeled off from the funnel in volumes. The strain quivered through every limb of the ship, but her captain kept her at it, and inch by inch she forced her way through the pounding seas.

“This manœuvre of the gallant British ship,” says an eye-witness, Mr. John P. Dunning, of the Associated U.S. Press, “is regarded as one of the most daring in naval annals. It was the one desperate chance offered her commander to save his vessel and the three hundred lives aboard. An accident to the machinery at this critical moment would have meant certain death to all. Every pound of steam which the _Calliope_ could possibly carry was crowded on, and down in the fire-rooms the men worked as they never had worked before. To clear the harbour, the _Calliope_ had to pass between the _Trenton_ (an American warship) and the reef, and it required the most skilful seamanship to avoid a collision with the _Trenton_, on the one hand, or total destruction upon the reef, on the other. The _Trenton’s_ fires had gone out by that time, and she lay helpless almost in the path of the _Calliope_.”

[Illustration: PROMENADE DECK OF THE “PARIS.”]

But the dreaded collision did not take place. And as the _Calliope_ passed near to the _Trenton_, a great shout was given for the British vessel, and the Englishmen responded with a noble cheer. Captain Kane, who subsequently was appointed to the _Inflexible_, said afterwards:

“Those ringing cheers of the American flag-ship pierced deep into my heart, and I shall ever remember that mighty outburst of fellow-feeling which, I felt, came from the bottom of the hearts of the gallant Admiral and his men. Every man on board the _Calliope_ felt as I did; it made us work to win. I can only say, ‘God bless America and her noble sailors!’”

The _Calliope_ did win. Her superb machinery and the fine seamanship with which she was handled were successful, and she returned to the harbour when the storm had subsided. Happily the brave men of the _Trenton_ also survived, though fourteen vessels were wrecked and nearly 150 lives were lost.

Strongly and staunchly as are built the Government ships, many of the great liners are their equals in these respects. Indeed, several of them are now retained by the Government to be used as armed cruisers should occasion require. The fittings and accommodation on many a large liner are also luxurious in the extreme. There are library and smoking-room, superb saloons and state-rooms, drawing-rooms,

music-rooms, and tea-rooms, bath-rooms, etc. In short, they are floating hotels of a most sumptuous character.

A modern steamship, with its multitude of comforts and conveniences for passengers and its complexities of machinery for fast and safe steaming, is a great triumph of engineering skill. Patience and forethought, the persevering development of sound principles, and the application of new ideas, have all contributed to this great achievement.

From the _Comet_ to the _Campania_ is a marvellous development within a century. And it has not been accomplished along one line, but upon many. The use of steel, of many-tubed and strong boilers, of high pressure steam, which would have frightened Henry Bell out of his senses, the forced draught and the surface condensers, the screw propeller, the direct-acting and the triple and quadruple expansion engines, have all contributed to the noble results. Steamships, with their complex, beautiful, and powerful machinery, may rank among the most wonderful things that mankind has ever made.

[Illustration]

[Illustration]

FAMOUS BRIDGES AND THEIR BUILDERS.

CHAPTER I.

“THE BRIDGE BY THE EARTHEN HOUSE.”

“You will not try again, surely?”

“Ay, I shall indeed!”

“What! after two failures?”

“Yes; I see the mistakes now. This bridge fell because it had too much weight on its haunches.”

“Haunches! you mean the two side-curves of the arch were too heavy.”

“Ay; you’ve heard the proverb no doubt that ‘An arch never sleeps.’ That is, should too great a weight fall on the crown or top part, the arch will fall at the sides outwardly, and the crown will sink; while, curiously enough, if it be built with too little weight on the crown, as this was, the crown will be forced upwards, and the sides will fall inwards.”

“Then you mean to build your third bridge with less weight proportionately on its haunches?”

“Exactly so.”

“Well, I wish you good luck, friend Edwards, for we need a bridge sorely over the brawling Taff.”

“You shall have it, neighbour. I shall succeed this time. I have gripped the right principle at last.”

He had indeed, for the bridge he then built lasts to this day. It was the famous Pontypridd bridge over the Taff on the Llantrissant and Merthyr road, and was called the Pont y du Prydd, or the bridge by the earthen house, for a mud hut stood near.

[Illustration:

PONTYPRIDD BRIDGE. _From Encyclopædia Britannica._]

About the year 1745 it was determined to build a bridge over the rushing Taff, and William Edwards, a self-taught mason of the country, undertook the task. The first bridge he built was of three arches, which, in less than three years, was dashed away by a great flood. The water rose so high as to surge over the parapet.

It must have been a sore disappointment to the hard worker to see his structure suddenly swept to ruins. But he was a shrewd, common-sense, observing man, and, nothing daunted, he tried again. This time he determined to build one bold arch of 140 feet. The object was to obviate the necessity of raising piers for more arches, and so obstructing the water; these former piers having caused, or assisted in causing, the destruction of his first bridge.

But the second gave way from the proportionally heavy weights on the haunches, as Edwards, we imagine, told his friend, and once more he had to face ruins. Yet a third time he tried, and the third time he was successful. Generations have come and gone, the children who played about its abutments have grown grey and have passed away, but still the country mason’s bridge of 140 feet span stands its ground and serves the community.

He reduced the heavy weight on the sides by making openings in the spandrels—that is, the part above the curve of the arch; while, instead of filling up the interior space with rubble, he used charcoal. But the arch is very steep, and a chain and drag is kept to assist any horse when descending.

These bridges illustrate the principle of the arch. Passing by the fact that it is evidently safer to span a swelling river by a bridge of wide, rather than of several narrow arches, three powers or forces act on the row of stones or bricks forming the arch. There is first the force that would carry the stone downward—that is, the force of its own weight and of anything that might be placed upon it. But then there are stones or bricks pressing against it on either side, and in its turn it presses upon them. When, therefore, every part presses equally, one not heavier or weaker than the others, a support for all is gained by the contiguous pressure and by the balance of forces.

Long bridges were sometimes built in this way, and the longest in England in the Middle Ages was at Burton, over the Trent. It was 1545 feet long, and had 36 arches. It was not superseded till 1864, when a new bridge was built.

In an arched bridge, the higher it rises in proportion to the width of the arch, the easier is its construction, and the less is the stress upon its parts; moreover, any

inaccuracy in design or in building is likely to be less harmful. We are not surprised, therefore, that Edwards, in his third attempt, decided upon that form.

One of the widest arches in the world is that of the famous Grosvenor Bridge at Chester. It has a span of 200 feet, with a rise of 42 feet. An arch, however, in the Washington Aqueduct extends to 220 feet span, while the central span in the Southwark Bridge, designed by Rennie, is 240 feet. This last, however, is of cast-iron.

The principle of the arch, however, does not appear first in the history of bridge building. Bridges are as old as mankind; that is, no one knows when first men began to cross streams and chasms by placing the trunk of a tree from one side to the other, and thus bridging the gulf.

Then, possibly, the next step was to build up a pile of stones in the centre of the stream—taking the stones there by coracle or canoe—and placing a tree trunk from the side to the central heap.

Yet another development would most likely be a simple cantilever bridge—though these early builders would not have known that Frenchified word. But they knew that after embedding a tree trunk firmly on each side of the bank so that a considerable portion should project over the stream, they could place a third log from one end to the other, and thus get a bridge much longer than when made of one tree trunk alone.

This principle, known so long ago, was used and immensely developed in the construction of the famous Forth Bridge, one of the most remarkable structures of the nineteenth century. This cantilever principle is very important in bridge building, and it is said that there exists an ancient bridge on this principle across the Sutlej in India with a span of 200 feet.

[Illustration: THE POST BRIDGE, DARTMOOR.

(_An example of an early bridge, of “slab” construction._)]

A further variety of early bridges was the “slab” bridge, consisting of slabs of granite placed from side to side, or from the sides of the bank to heaps of stones piled up in the stream. A good example of such a bridge may be seen at “Post Bridge” over the Dart on Dartmoor. Ages ago this bridge was built, and as we study it and compare it with the modern structure not far distant, we wonder how the ancient Britons—if those sturdy individuals are really responsible for it—could raise and place those huge slabs of stone without engineering apparatus. Probably it was done with levers and rollers, and there must have been many shoulders to the wheel in the process. Certainly they had plenty of granite at hand on wild Dartmoor.

But passing by all these early forms of bridges—which it will be noticed are built of a few large pieces of material—it was left to the Romans, at all events in Europe, to largely adopt the arch as a principle of construction.

Now, here we are dealing with an altogether different principle. The arch is made up of a number of comparatively small pieces of material bound together by mortar, or cement, or even clamps, and by the power of gravitation.

We doubt if that idea is realised by half the people using the multitudinous arches abounding to-day; yet it is true. Or to put it in another way, the various parts are arranged so that they keep up each other by pressure.

If you take two cards, or bricks, or slabs of stone and lean them together at the top, while the other ends may be far apart, you will find they will bear a certain amount of weight. Here you have the principle of the arch in its simplest form; and it may be that out of that primitive performance the arch has grown. This kind of triangular arch is to be met with in ancient structures in Great Britain. The flanks or haunches of an arch are its sides, from the first stone to the keystone; and the crown is its highest part; while the central wedge-shaped piece of stone or brick is called the keystone.

The stones or bricks are cemented together when being built over a framework of timber, called the centering, and when the keystone is placed and the arch is complete it ought to remain firm.

But should too great a weight fall on the crown the bridge will fall outwardly at the sides, and the crown will sink; while, curiously enough, if it be built with too little weight on the crown, it will be, as it were, forced upwards, and the sides will fall inwards, as in the case of the second of the famous Pontypridd bridges, which actually did this. The material in the middle of the arch was less in proportion than that over the sides or “haunches,” and these heavier weights on the sides caused the crown to be forced upwards.

Two causes combined to make changes in bridge building. These were the needs of railways and the introduction of iron as a building material. The first iron bridge was constructed over the Severn, near an appropriately named place, Ironbridge, in 1779. It had an arch of near upon a hundred feet span.

When, however, very wide span bridges were required, the question arose of the superiority of wrought-iron over cast-iron for such structures. The Menai Strait had to be crossed for the Chester and Holyhead Railway, and the greatest existing cast-iron span was Rennie’s Southwark Bridge, where 240 feet had been reached. But over the Conway and the Menai Strait, spans of 400 feet were involved. How were these yawning gulfs to be bridged?

CHAPTER II.

A NEW IDEA—THE BRITANNIA TUBULAR.

“We must cross the Strait at the Britannia Rock—that is settled.”

“And where is the Britannia Rock?”

“Nearly in mid-channel. It seems placed there for the purpose.”

And the great engineer smiled.

“What are the distances?”

“From coast to coast the span of the Strait is some 1100 feet, with that rock in the centre. Now the problem is, to build a bridge across that gulf of surging water strong enough to bear heavy trains at high speeds, and sufficiently above the water to prevent any interference with navigation.”

“And how will you manage it?”

“First I thought of large cast-iron arches, but they will not do. I doubt if they would stand the strain; and moreover we should impede navigation by raising scaffolding during the building. At length I came to the idea of a tube bridge.”

“What! a tube bridge! I’ve never heard of it!”

“No, it is a new idea. By reconsidering a design I had made for a small bridge over the Lea at Ware in 1841, and thinking over the matter, I came to the idea that a bridge consisting of a hollow beam or tube might solve the difficulty.”

“A huge hollow girder, so to speak!” exclaimed his friend.

“Exactly so. Accordingly,” the engineer continued, “I had drawings prepared and calculations made, by which to ascertain the strength of such a bridge, and they were so satisfactory that I decided on attempting one.”

“It is like constructing one huge hollow beam of iron by rivetting plates together. Can it be done?” remarked his friend.

“The making of the high-level bridge over the Tyne, in which I had a part—the bridge between Newcastle and Gateshead, you know—was a transition between an arched bridge and a girder bridge. A girder of course is a beam, it may be of iron or wood, and the little bridge at Ware has been built of girders made of plates of wrought-iron rivetted together. Therefore, you see, I am not unused to wrought-iron girders, and what they will bear.”

“Why, it is like a huge extension of the primitive log-bridge of our ancestors.”

“If you like,” replied the engineer, laughing.

Robert Stephenson—for he it is whom we suppose to be speaking to his friend on this gigantic engineering enterprise—became satisfied by reflection that the principles involved in constructing an immense tubular beam were but a development of those commonly in use; and Sir William Fairbairn was entrusted with the duty of experimenting as to the strength of tubes, the directors of the Railway Company voting a sum of money for the purpose.

Sir William, then Mr., Fairbairn concluded that rectangular tubes were the strongest, and a model was made of the suggested bridge. It proved successful, and indicated that the tube would be able to stand the strain of a heavy train passing rapidly over it.

In September, 1846, Mr. Fairbairn read a paper on the subject at the meeting of the British Association at Southampton, as also did Professor Hodgkinson, a mathematician, who had verified Fairbairn’s experiments. Not long afterwards Stephenson became satisfied that chains were not needed to assist in supporting the

bridge, and that his tubes would be strong enough to support themselves entirely between the piers.

Work therefore went forward. Some 1500 men were engaged on the Britannia Bridge, and the quiet shores of the Menai Straits resounded with the busy hum of hammers and machinery. Cottages of wood were built for the men, and workshops for the punching and rivetting of the plates for the gigantic tubes.

The design included two abutments of masonry on either side of the Strait, and three towers or huge piers, one of which, the centre pier, was to rise from the Britannia Rock, 230 feet high. There are four spans, two over the water of 460 feet each, and two of 230 feet each over the land. Two tubes, quite independent of each other, but lying side by side, form the bridge across. Each tube or beam is 1510 feet long, and weighs 4680 tons. Its weight at one of the long spans is 1587 tons.

Now how could these gigantic tubes be put together and raised to their positions? Here was a problem almost as great as the original one of the bridge itself, and it troubled the engineer sorely.

[Illustration: ROBERT STEPHENSON.]

“Often at night,” he declared, “I would lie tossing about, seeking sleep in vain. The tubes filled my head. I went to bed with them, and got up with them. In the gray of the morning, when I looked across Gloucester Square, it seemed an immense distance across to the houses on the opposite side. It was nearly the same length as the span of my tubular bridge.”

The principle adopted was to construct the shorter tubes on scaffolds in the places which they were to occupy. This could be done, for such scaffolding would not impede navigation. But scaffolding could not be built for the large tubes across the great spans of water. What then was to be done?

It was decided to build them on platforms on the shore quite close to the water, and float them when ready on pontoons to their places between the piers, raising them to their position by hydraulic power. Such a task would be hazardous enough. It was first tried at Conway, where a similar bridge was being built by Robert Stephenson, being indeed part of the same railway. The Britannia was, however, a much greater enterprise, though the span of the Conway is 400 feet. The Conway bridge, indeed, is but of one span, and contains two tubes.

The experience at Conway was of great benefit to the gigantic undertaking at the Menai Strait. The floating of the first tube was to take place on the 19th of June, 1849, in the evening; but owing to some of the machinery having given way, the great event was put off to the next night. The shores were crowded with spectators. When the tube was finished it could be transferred to the pontoons; for the tubes had been built at high-water mark. When the pontoons were fairly afloat on this fateful evening, they were held and guided by leading strings of mighty strength. Stephenson himself directed in person, from a point of vantage at the roof of the tube. Thence he gave the signals which had been agreed upon, whilst a crew of sailors, directed by Captain Claxton, manned the strange barque.

A pontoon is a light, buoyant boat, and the tube was supported on sets of these, their speed increasing terribly as they approached their place by the towers. The idea was, as related by Mr. Edwin Clark, Stephenson's assistant, that they should strike a "butt" properly, underneath the Anglesey Tower, "on which, as upon a centre, the tube was to be veered round into its position across the opening. This position was determined by a twelve-inch line, which was to be paid out to a fixed mark from the Llanfair capstan. The coils of the rope unfortunately over-rode each other upon this capstan, so that it could not be paid out."

Destruction seemed imminent. The capstan was actually dragged from the platform, and the tube seemed likely to be swept away. Then Mr. Rolfe, the captain of the capstan, shouted to the spectators, and threw out a spare twelve-inch rope. Seizing this, the crowd, with right good-will, rushed it up the field, and clung tightly to it, checking the voyage of the mighty tube. It was brought to the "butt," and duly turned round.

A recess had been left in the masonry of the tower, and the end near the Britannia pier was drawn into it by means of a chain. The Anglesey end followed. Then the tide gradually sank, the pontoons sank with it, and the tube subsided also to a shelf which had been made at either end. The first stage was accomplished; the mighty tube was in position to be raised.

Shouts of rejoicing burst from the sympathetic crowds, and the boom of cannon joined its congratulatory note at the grand success. But the further stages remained. At midnight the pontoons were all cleared away, and the huge, hollow beam hung silent over the surging water. It rested on the shelves or beds prepared for it at either end. The second great operation, of course, was to haul it up the towers to its permanent position. This was to be performed by hydraulic machinery of great power, and Mr. Stephenson's instructions were to raise it a short distance at a time, and then build under it.

He took every imaginable precaution against accident or failure; and well was it that he did so, for an accident happened which, but for the careful building under the tube in the towers as it was raised, would have been most calamitous. The accident occurred while Mr. Stephenson was absent in London. One day, suddenly, while the machinery was at work raising the tube, the bottom burst from one of the hydraulic presses, and down fell the tube on to the bed provided for it.

Though the fall was but nine inches, tons weight of metal castings were crushed, and the mighty tube itself was strained and slightly bent. But it was serviceable still, and the fact that it stood the strain so well showed its great strength. It weighed some five thousand tons, and for such an immense weight to fall even three-quarters of a foot was a very severe test.

But for Stephenson's wise precaution in lifting it slowly, and building underneath it as it was raised, the tube would have crashed to the bottom of the water. As it was, the accident cost £5000; but the tube was soon being hauled upward again. In due course the others followed, and on the 5th of March, 1850, Robert Stephenson inserted the final rivet in the last tube, and the bridge was complete. He crossed over

with about a thousand persons, three locomotives whirling them along.

The tubes of the bridge are made of iron plates, and at the top and bottom are a number of small cells or tubes—instead of thick iron plating—which assist in giving strength to the whole gigantic tube. Thus it may be said the floor and roof are tubular, as well as the body. These hollow cells appear to have been Fairbairn's invention. The size of the tube grows slightly larger at the middle by the Britannia tower, where externally the tubes are 30 feet high, and 26 internally, while they are 22¾ feet and 18¾ feet at the abutments. The width is 14 feet, 8 inches externally, and 13 feet 5 inches inside.

At the Britannia tower the tubes are placed solidly on their bed, but at the abutments, and at the land towers, the tubes rest on roller-beds. This arrangement was adopted to permit of expansion and contraction. Iron, of course, solid and unyielding as it appears, is yet very susceptible to warmth, and the effect of the sun's rays on this massive iron structure is very marked. A rise of temperature causes it to expand in a comparatively short time, and it is said that the tubes occasionally move two and a-half inches as the sun gleams upon them. Mr. Edwin Clark observed the effect of the sun on the iron, which appears in a small degree to be always moving as the temperature varies. Well, therefore, that the able engineer planned an arrangement allowing for this constant expansion and contraction of the iron mass.

[Illustration: THE BRITANNIA TUBULAR BRIDGE.]

The Britannia Bridge was a great triumph for Robert Stephenson. He appears first to have seized the idea, and, assisted no doubt by Fairbairn's experiments and by able coadjutors, he carried it through to a successful completion. He was of course the son of George Stephenson, who had done so much for the locomotive, and according to Smiles, "he almost worshipped his father's memory, and was ever ready to attribute to him the chief merit of his own achievements as an engineer."

"It was his thorough training," Mr. Smiles once heard him remark, "his example, and his character, which made me the man I am." Further, in an address as President of the Institution of Civil Engineers, in January, 1856, he said: "All I know, and all I have done is primarily due to the parent whose memory I cherish and revere."

That father had died before the Britannia Bridge was completed, though he had been present at the floating of the first tube at Conway. The great engineer passed away on the 12th of August, 1848, at the age of sixty-seven, and his distinguished son Robert, who had no children, only survived him by eleven years.

But before he died he had designed, and Mr. A. M. Ross, who had assisted at the Conway Bridge, had assisted in carrying out the celebrated Victoria Tubular Bridge over the great St. Lawrence River at Montreal.

This bridge was for the Grand Trunk Railway of Canada, and for immense length and vastness of proportions, combined with magnificent strength, is one of the wonders of the world. It is five times as long as the Britannia Bridge, being not far short of two miles. It has a big central span of 330 feet, and twenty-four spans of 242 feet. The iron tubes are suspended sixty feet above the water beneath.

[Illustration: VICTORIA TUBULAR BRIDGE, MONTREAL.]

One great difficulty in the problem was the ice. Immense quantities come down in the spring, and to resist this enormous pressure the piers are most massive, containing thousands of tons each of solid masonry. These piers are based on the solid rock, the two central towers being eighteen feet in width and the others fifteen feet. To protect them from the ice, huge guards made of stone blocks clamped with rivets built up in the form of an incline were placed before the piers on the up-stream side. The bridge was begun in July, 1854, and occupied four and a-half years in construction, it being completed in December, 1859, about two months after its designer had died.

Gigantic though this structure is, and great as is the honour which it reflects on Robert Stephenson and the resident and joint engineer Mr. Ross, yet with the exception of the remarkable and massive ice-guards to the piers, it does not differ materially from the Britannia and Conway Tubular Bridges. These were the first famous examples of the new principle.

Why, then, are massive tubular bridges not more generally built? Because they led to another and very natural development in bridge-building, a development whereby great strength for long spans is gained, with, however, a marked saving both in labour and in material. That development was the lattice bridge.

CHAPTER III.

LATTICE AND SUSPENSION BRIDGES.

“The expense of a tubular bridge would be too great.”

“But if we could get the strength without the expense.”

“What mean you?”

“By iron lattice work we could, I think, gain the stiffness and support needed, without such great cost of labour and material. In other words, I propose a lattice or trellis work girder, instead of a solid sided, or a tubular girder.”

“That is, you would have the sides of lattice or trellis work, instead of solid plates?”

“Exactly. I would use bars of iron placed diagonally. These lattice or trellis bridges are developed from the tubular bridges, also from the loose wooden lattice bridges of America. We make a web of iron instead of a solid sheet. The same kind of structures are largely used over the wide rivers of India. Sir John MacNeill designed the first in iron, and it was built in 1843 on the Dublin and Drogheda Railway with a span of eighty-four feet. I consider they will be among the most popular bridges of the future for longish spans.”

The engineer’s prediction has come true; for lattice bridges have undoubtedly been very widely adopted. We may suppose that he was advising the directors of a proposed railway, and we doubt not but that he carried the day.

A fine specimen of a lattice bridge is that across the Thames near Charing Cross, for the South-Eastern Railway. It has a total length of more than a quarter of a mile—viz., 1365 feet, and six of its nine spans are 154 feet wide. Two principal girders, fourteen feet deep, are connected transversely by other girders which carry the rails and project on the other side to support a footpath. The two main girders are nearly fifty feet apart and one weighs 190 tons.

The sides have upper and lower booms made of plate iron connected by perpendicular bars, between which are a couple of bars crossing each other diagonally at an angle of forty-five degrees, and fixed to the booms by bolts of five and seven inches in diameter.

The old Hungerford Bridge stood here previously, and its two piers of brickwork were used for the new bridge. Other piers are huge cylinders of cast iron ten feet across, but fourteen feet in diameter in the ground. Thus they are broadly based. These piers are filled with concrete and also brickwork, and are topped with bearing-blocks of granite. They are formed of plates of cast iron bolted together, and they were sunk into the ground many feet below high-water by combined forces; divers scooped out the mud and gravel and clay from within the cylinders; water was pumped out and heavy weights pressed them down. The piers became fixed on the London clay, but when filled were heavily weighted to drive them down again, and finally they were forced to a depth of over sixty-two feet below high-water mark.

But before lattice girder bridges had become so popular, another class had come into use, and afford some splendid specimens of engineering skill. These are suspension bridges, and, perhaps of all kinds, they are the most picturesque. Their graceful sweeps and curves yield perhaps a more pleasing sight for the eye than the solid, rigid, straight lines of the girder bridges.

It was the genius of Thomas Telford which gave a great impetus to this class of bridge. Like Stephenson after him, he had to bridge the surging Menai Straits, but for a carriage road, not a line of rails; and at length, after various plans had been suggested and abandoned, he proposed the Suspension Bridge.

Now, in its simplest form, a suspension bridge has been known for ages. It is merely a pathway, or even a small movable car, suspended from a rope or ropes across a chasm. Ulloa describes suspension bridges built by the Peruvians in South America. Four stout cables span a river, and on these four is placed the platform of sticks and branches, while two other ropes connected with the platform are useful as hand rails. Such bridges sway with the wind and move with the passenger, but for light loads they appear to be perfectly safe.

In Telford’s Menai Bridge the carriage-way is hung from four huge chains or cables, each chain made up of four others, and passing over high piers. The chains are anchored on the landward side, sixty feet in pits, and grafted by iron frames to the rocks. The chains are so complex and so strong, that parts may be removed for repair without imperilling the safety of the structure. The length of the span thus gained is 560 feet, and it is 150 feet above high-water. The remainder of the bridge is composed of arches of stone, of 52½ feet span.

The piers from which the great span is suspended rise above the carriage-way fifty-two feet, and are topped by blocks of cast-iron, which can move on rollers to permit the chains passing over them to expand and contract freely with the temperature. There are two carriage-roads, and also a footpath. The roads are separated by iron lattice work, which also gives them stability and decreases vibration.

[Illustration: THE CLIFTON BRIDGE.]

In its day, this stupendous bridge was as great a wonder as its later companion over the same Straits—the Britannia Tubular. Six years were occupied in building, and it was opened in 1825. Why, then, did not Stephenson construct a similar bridge when, twenty years or so later, he had to solve a similar problem?

The answer is, that suspension bridges are not—or were not—considered sufficiently strong and rigid for railway work. In America, however, they have been used for this purpose; witness the famous Niagara Suspension Bridge, 2⅓ miles below the Falls, and with a superb span of 822 feet; but American engineers appear to stiffen the roadway considerably, so as to distribute the stress of the rushing train over a large portion of the cable. The Niagara Bridge is not supported by plate-link chains, but by four immense wire cables, stretching from cliff to cliff over the roaring rapids. Four thousand distinct wires make up each cable, which pass over lofty piers, and from them hangs the railway by numerous rods.

[Illustration: THE BROOKLYN BRIDGE.]

Probably the famous Brooklyn Bridge is the largest suspension bridge in the world, even as the Clifton Suspension Bridge, in England, is one of the most interesting. The Brooklyn Bridge has a magnificent central span of 1595½ feet over the East River between Brooklyn and New York; further, there are two land spans of 930 feet, which, together with the approaches, make up a total of about a mile and a furlong. The cables, four in number, are each composed of 5000 steel wires, and measure 15¾ inches in diameter. They are anchored to solid stone structures at either end, measuring 119 feet by 132 feet, and weighing 60,000 tons; while the towers from which the main span is suspended rise to the height of 276 feet, and are embedded in the ground 80 feet below high-water. It has been estimated that the weight hung between these towers is nearly 7000 tons.

The roadway of the bridge is divided into five thoroughfares. Those on the outer sides are for vehicles, and are 19 feet wide; the centre is for foot passengers, and is 15½ feet in width; while the two others are for tramway traffic. The bridge was opened in 1883, and affords a great triumph of engineering skill.

Much smaller, but none the less interesting, is the Suspension Bridge at Clifton. As far back as 1753, Alderman William Vick, of Bristol, left a sum of £1000 to build a bridge at Clifton. The sum was to lie at compound interest until £10,000 was reached. However, the money was increased by subscriptions, and in 1830 an Act of Parliament was obtained for its construction.

The work coming into the hands of Mr. I. K. Brunel, he designed a bridge of 702 feet span, and 250 feet above high-water. The piers and abutments were built, but lack of cash, which forms an obstacle to so many brilliant enterprises, stopped the progress of the bridge for nearly fourteen years.

Then it occurred that the Hungerford Suspension Bridge was to be removed to make way for the Charing Cross Railway Bridge, so the chains were purchased at a comparatively small cost, and the work at Clifton proceeded, and was finally completed.

Three chains on either side suspend long wrought-iron girders, which help to stiffen the platform; and cross girders between support the floor. The chains pass over rollers on the piers, and are ultimately anchored to plates bedded in brickwork abutting on rock. The platform is hung by upright rods from the chains, and hand-railing is used with lattice-work, to assist in rendering it rigid. The roadway, twenty feet wide, is made of creosoted wood, five inches thick, while the pathways on either side are made with wood half as thick. Between the piers the weight of the structure, including the chains, amounts to nearly a thousand tons.

In all these suspension bridges, however large, the principles are much the same. The platform, or roadway, is hung from chains or cables, which pass over piers and are anchored fast at the ends. Some are stiffened with girders and bracing to prevent undue undulation. The chains take a graceful and definite curve, that of the Menai Bridge dipping fifty-seven feet. The strain is the greatest at the lower part, and is increased, should the chain be drawn flatter over the same space. These bridges became widely adopted.

But there came a time when none of the bridges in vogue seemed to give what was required. A new principle was wanted. Where was it to be found?

CHAPTER IV.

THE GREATEST BRIDGE IN THE WORLD.

“Have you heard the news? The Tay Bridge is blown down!”

“Yes. A terrible disaster. I should think they would give up their scheme of bridging the Firth of Forth after that.”

“Not they! The scheme may be altered, but bridge it they will. Engineers never give in.”

The comments of these newspaper readers were right. The Tay Bridge, the longest in the world, had been blown down one wild December night in 1879, and girders, towers, and the train which was rushing over it, were suddenly hurled into the surging flood.

At that time a scheme was in hand to bridge the Forth for the North British Railway system, and Sir Thomas Bouch had proposed two suspension bridges hung by steel chains. But ultimately a new design altogether was adopted, the plan being by Sir Benjamin Baker and Sir John Fowler.

It was the new principle—or, rather, a remarkable development of an old principle—for which the bridge-making world was waiting: the principle, namely, of the cantilever.

A cantilever is, in fact, a bracket; and Sir Benjamin Baker has described it as such. It is a strong support, built out from a firm base, and is like a powerful and magnified bracket upholding a shelf.

In the Forth Bridge there are two huge spans, 1700 feet wide, crossed by these cantilevers; bridging channels of some 200 feet deep.

The longest spans on the Tay Bridge were 245 feet; it was over two miles long, and had ninety spans. It was an iron girder bridge, and was opened on the 31st of May, 1878. Not to be beaten, however, after the panic had subsided, another and more stable bridge was constructed, also a girder, but not so high in elevation, and sixty feet further up the river. It was opened in 1887, and is 10,779 feet long, with 85 piers, the navigable channel being under four of the spans, the centre spans being 245 feet wide.

It will be seen at once that the cantilevers at the Forth Bridge cover very much wider spans; and the channel being so deep, the impossibility of building piers will also be obvious. The best place for the bridge was marked by the projection of the Inverkeithing peninsula on the north shore, and also the Inchgarvie rock in the channel itself. The peninsula brought the two shores together, reducing the space to be bridged, and the rock gave firm support for a pier. Still there were the two immense spans of 1700 feet to be crossed, and the engineers decided on the cantilever principle. Thus, though the Tay Bridge was the longest in the world, the Forth presented by far the greatest spans—viz., the two main spans of 1700 feet each, in addition to which there are two of 675 feet each, and fifteen of 168 feet each.

The total length of this magnificent bridge, which Sir Benjamin Baker rightly claimed was the most wonderful in the world, is somewhat over 1½ miles in length, or 8296 feet, including the piers, while almost a mile is bridged by the huge and superb cantilevers. This is, perhaps, the great marvel. The clear space under the centre is no less than 152 feet at high-water, while the highest portion is 361 feet above the same mark.

And now, how was this great bridge constructed? Workshops were erected at South Queensferry, and the mammoth cantilevers were put up there piece by piece. They were fitted together and then taken plate by plate to the bridge itself. The shops were lit by electricity, and furnished with appliances for bending, cutting, moulding, holing, and planing plates. The workshops were surrounded by quite a maze of railways.

But what of the piers, without which all these preparations would be unavailing? Now the foundations of piers are usually laid by means of cofferdams; that is, piles of timber are driven down through the water into the bed of the river close together, and the interstices filled with clay; or a casing of iron may be used instead. The water in the enclosure thus formed can be pumped out and excavation proceeded with, and the foundations laid. Cofferdams are sometimes made of iron boxes or caissons with

interstices fitted with felt, and caissons of this kind about 12½ feet long and 7 feet wide were used in constructing the Victoria Embankment on the Thames.

But with certain of the piers for the Forth Bridge the water was too deep for timber cofferdams, and the usual diving-bell was not sufficiently large. The piers were to be of immense size, no less than 55 feet in diameter, and the diving-bell of ordinary size would not cover that great width.

Huge caissons were therefore made, 70 feet wide, constructed of iron plates and rising in height, according to the depth of water, up to 150 feet. The lower part of the immense caisson or tank was fitted as a water-tight division and filled with compressed air, the object being to resist the pressure of the water. Two shafts communicated with this air-tight division or mining chamber, one for the removal of the earth excavated, and the other for the men to pass up and down. The escape of the air through the shafts was prevented by the use of an air-lock, working on the same principle as a water-lock on rivers or canals. There were two doors in the lock, one communicating with the shaft and the other with the outside air. When the latter was closed and the lock filled with compressed air by opening a valve or tap, the door of the shaft could be opened and the man could descend to his work below.

That work consisted chiefly of excavation in the bed of the river. Drills, hydraulic cutters, and dynamite blasting were all utilised until huge holes, many feet below the river bed, were hollowed out. As the caisson was filled with concrete above the air-tight chamber where the men worked it was exceedingly heavy, and sank by its own weight into the space prepared.

The mining chamber was lit by electricity, and was about seven feet high. The mud of the river bed was mixed with water and blown away by the compressed air which seems to have been about 33 lbs. to the square inch. The caissons were sunk down to rock or boulder clay, and when they had reached the required distance the mining chamber was filled with concrete, and the same material used to the level of the water; the piers were then built up with huge stones placed in cement, the whole forming a magnificent mass of concrete and masonry, carried down in some cases to about 40 feet below the bed of the river.

[Illustration: THE FORTH BRIDGE.]

The three chief piers consist of groups of four columns of masonry, each gradually tapering from 55 feet in diameter to 49 feet at the top, and about 36 feet high. From these rise the huge cantilevers connected together by girders 350 feet in length.

The centre of these three main piers rests on the island of Inchgarvie; the two others are known as the Fife and the Queensferry piers respectively, and are placed on the side of the deep water channels. In addition to these three main piers are several others, some in shallow water and some on land. The part of the bridge which they carry is an ordinary girder of steel leading to the immense cantilevers. For founding the shallow water piers, cofferdams were used; the caissons with compressed air chambers being for the deep water structures.

They were put together on shore, launched, floated, steered to the desired position, and sunk. One proved cranky and turned over, and was only brought right after much expense and difficulty.

The cantilevers are bolted down to each pier by numbers of huge steel ties, 24 feet in length and 2½ inches in diameter, embedded in the masonry, there being 48 of these bolts or ties to each column. And now as to these cantilevers.

Four huge tubular shafts, two on each side, rise from the group of columns forming each pier, to the height of 350 feet. From these shafts, which slope slightly inward, project the cantilevers, the upper and lower parts being strongly braced together by diagonal ties. In shape the gigantic brackets taper towards a point, the width decreasing as much as from 120 feet at the commencement of the piers to 32 feet at the ends. The wind, it is believed, will be more effectually resisted by this means.

The cantilevers are hung back to back, one to some extent counter-weighing the other. The component parts consist of cylinders of steel or struts for resisting compression—these are the lower parts; and ties of lattice-work made of steel plates for resisting tension,—placed above.

Thus, then, from each of the three chief piers two pairs of gigantic brackets project, each pair placed side by side and braced together, and forming one composite cantilever jutting to the north and one to the south. The rails run on sleepers placed lengthwise and fixed in troughs of steel, so that should a train run off the line the wheels will be caught by these supports.

It is calculated that there are about 45,000 tons of steel in the bridge, and 120,000 cubic yards of masonry in the piers. The contract price was £1,600,000, which works out at about £215 per foot; and the contractors, who were able to obtain an admirable organisation of some 2000 men to carry out the magnificent design, were Messrs. Tancred, Arrol, & Co. Some special tools for use in the work were planned by Sir William Arrol. The bridge was opened by the Prince of Wales on the 4th of March, 1890.

The success of this magnificent structure has assured the wider adoption of the cantilever principle. Long-span bridges, in several cases, have since been built on this design. Its engineers may claim indeed to have widened the scope and possibilities of bridge-building.

Still, when another bridge was wanted over the Thames, at a busy spot, crowded with shipping and near the historic Tower of London, another kind of structure was adopted. What was it?

CHAPTER V.

THE TOWER BRIDGE.

“Why should they not have a drawbridge?”

“What! To draw up from each bank of the river?”

“No, I did not mean that exactly. Could they not get piers farther in towards the centre of the stream, and let the drawbridge rise and fall from them?”

“The river is too crowded for many piers.”

“It is. But I cannot help thinking a drawbridge—a bascule bridge as the engineers call it—is the best solution of the difficulty.”

“Well, a bridge is wanted sufficiently low to spring from the flat banks of the Thames for foot passengers and carriage traffic, and yet sufficiently high to permit tall ships to pass underneath.”

“And apparently these two requirements are incompatible.”

“Not altogether,” remarks a third speaker.

“You are partly right in your idea of a drawbridge. That is Sir Horace Jones’s idea. And, further, there is literally to be a high and also a low-level bridge; for there are to be two levels—that is, two roadways—one at a high, and one at a low, level across the middle span.”

“And is the low level to be a drawbridge—a roadway that can be drawn up to permit vessels to pass? Is that so?”

“Exactly. And this drawbridge will be in two parts, one on either side; they will be worked from two massive piers giving a clear span of 200 feet in the middle of the stream, through which span big vessels can pass. The usual traffic of the river will be able to pass even when the drawbridges are down.”

“And above the bascules or drawbridges will run the high-level bridge?”

“Yes, a girder bridge for footpaths, and people will reach it by lifts and staircases in the piers—which, by-the-by, will be more like huge towers. These towers will also contain the machinery for raising and lowering the drawbridges.”

“And what sort of bridge will be used for the other spans—that is, to cross the river between the piers and the shore?”

“Suspension bridges; so that the Tower Bridge as it will be called, for it will cross the Thames by the Tower of London, will embody the suspension, the bascule (or drawbridge), and the girder bridge principles, while in the centre will be two levels.”

“It promises to be a splendid piece of work.”

“It does. And it is very much needed, for the congestion of traffic on London Bridge is terrible.”

“And people have often to come round a long way to reach it.”

The promise of the Tower Bridge, as set forth by these speakers, has been amply fulfilled. It is indeed a fine piece of work; and although it does not embody any new idea, yet in its combination and development of old principles and in its size it is very remarkable. It was opened in June, 1894, and is, or was at the time of building, the biggest bascule bridge in the world.

Within its handsome Gothic towers are steel columns of immense strength, constituting the chief supports of the suspension bridges and of the high-level footways. The architect was the late Sir Horace Jones, and the engineer Mr. J. Wolfe Barry, while the cost was, including land, about £1,170,000.

The problem was to combine a low-level bridge providing for ordinary town traffic with a high level, under which ships could pass, and it was accomplished by a union of principles. In its oldest shape the drawbridge was probably a huge piece of timber, which was hauled up and let down by chains over the moats of castles. In the Tower Bridge there are two of such huge “flaps” or leaves, each about 100 feet long, one rising and falling from each pier and meeting in the centre. Large bascule bridges are usually constructed in this manner, and there is an excellent specimen over the Ouse, for the passage of the North-Eastern railway; one man at each half of the bridge can raise it in less than two minutes. Another fine bascule may be seen at Copenhagen.

The bascules are raised and lowered by chains, which, in the case of the Tower Bridge, are worked by superb hydraulic power from the massive pier towers. When drawn up, which is done in less than five minutes, the bascules are even with the sides of the towers, and full space is given for the vessels to pass.

The two side spans of the bridge, crossed by the suspension bridges, are wider than the centre, being 270 feet each, and the total length of the whole bridge is 800 feet between the abutments. There are also piers on the shoreward side for carrying the chains of the suspension bridges at each extremity.

The massive tower piers, sunk 27 feet below the river bed, are built of gray granite, and are also fitted with strong break-waters to resist the action of the tide. The high-level bridges across the central span are for foot passengers, and are 135 feet over high-water mark. The bascule bridges, when closed for vehicular traffic, are 29½ feet above high water, while the side suspension spans are 27 feet. The roadway is 50 feet wide, which is also the width of the approaches. The foot passenger traffic is never stopped, as persons can pass by the hydraulic lifts or the stairways in the tower piers to the high-level bridges above.

Sir Horace Jones died before the great work was completed, and was succeeded by Mr. G. D. Stevenson, who had been his assistant. Sir William Arrol & Co. supplied the iron and steel, and Sir William Armstrong the hydraulic machinery. Various contractors carried out different portions of the mighty work, which occupied about eight years in building. Near by stands the ancient Tower of London, looking not unkindly on the great constructive effort to which it has given its name.

Sometimes a bridge is made movable by swinging it round on a pivot instead of drawing it up on a hinge or axis; and sometimes, as in the case of a bridge over the Arun for the Brighton and South Coast Railway, it is made to slide on wheels backwards and forwards from the abutment. Floating or pontoon bridges are made by placing planks on pontoons, or boats anchored by cables. The longest in the world is probably at Calcutta, across the Hooghly. It is 1530 feet in length, there being twenty-eight pontoons in pairs. These are of iron, 160 feet long, and with ends shaped like wedges; they support a road-way of 3-inch timbers, forty-eight feet wide, and

raised on tressel work. An opening can be made for ships by removing four pontoons and floating them clear of the passage way.

Great bridges present some of the most remarkable triumphs of the engineer. They rank beside the express locomotive and the ocean liner as among the great constructive achievements of mankind. Daring in design, and bold in execution and in sweep of span, they have been developed along several principles; and so solidly have they been built, so sound are the laws of their being, that it seems as though they will live as long as the everlasting hills.

[Illustration]

[Illustration]

REMARKABLE TUNNELS AND THEIR CONSTRUCTION.

CHAPTER I.

HOW BRUNEL MADE A BORING-SHIELD.

“I watched the worm at work and took my idea from that tiny creature!”

“A worm! Was it an ordinary worm?”

“Oh no, it was the naval wood-worm—_Teredo Navalis_; it can bore its way through the hardest timber. I was in a dockyard and I saw the movements of this animal as it cut its way through the wood, and the idea struck me that I could produce some machine of the kind for successful tunnelling.”

“Well, it has been brilliantly successful.”

“I looked at the animal closely, and found that it was covered with a couple of valvular shells in front; these shells seem to act as a shield, and after many attempts I elaborated the boring-shield which was used in hollowing out the Thames Tunnel.”

This statement, which we can imagine to have been made by Sir Marc Isambard Brunel to a friend, is no doubt in substance quite true. A writer in the “Edinburgh Encyclopædia” says, that Sir M. I. Brunel informed him, “that the idea upon which his new plan of tunnelling is founded, was suggested to him by the operations of the _Teredo_, a testaceous worm, covered with a cylindrical shell, which eats its way through the hardest wood.”

Two or three attempts had already been made to drive a tunnel under the Thames, but they had ended in failure. In 1823, Brunel came forward with another proposal, and he ultimately succeeded.

This illustrious engineer must not be confounded with his son—who was also a celebrated engineer—Isambard Kingdom Brunel. There were two Brunels, father and son, even as there were two Stephensons, George and Robert.

Sir Marc Isambard Brunel, the father, whose most notable enterprise was the Thames Tunnel, was a French farmer’s son, and after various experiences in France and America settled in England in 1799, and married the daughter of William

Kingdom of Plymouth. He had already succeeded as an engineer so well as to be appointed chief engineer of New York, and a scheme for manufacturing block-pulleys by machinery for vessels was accepted by the British Government, who paid him £17,000 for the invention. He was also engaged in the construction of Woolwich Arsenal and Chatham Dockyard, etc., and in 1823 he came forward with another proposal for the Thames Tunnel.

In that same year, his son, Isambard Kingdom Brunel, entered his father's office, and assisted in the construction of the tunnel. The son subsequently became engineer to the Great Western Railway, and designed the _Great Western_ steamship.

But though Brunel's proposal for the tunnel was made public in 1823, the work was not actually commenced until March, 1825. It was to cross under the river from Wapping to Rotherhithe, and present two archways. And if you had been down by the Rotherhithe bank of the Thames about the latter date, you would have been surprised to see that instead of hollowing out a shaft, proceedings began by raising a round tower.

A space was traced out, some 50 feet across, and bricklayers began to build a circular hollow tower about 3 feet thick and 42 feet high.

This tower was strengthened by iron bars, etc., and then the excavation commenced within. The soil was dug out and raised by an engine at the top, which also pumped out water. And as the hollow proceeded, the great shaft or tube of masonry sank gradually into it. Bricklayers added to its summit until it reached a total height of 65 feet, which in due course was sunk into the ground.

Thus, then, the engineer had, to commence with, a strong and reliable brickwork shaft, 3 feet thick, by which men and materials could ascend and descend in safety. A smaller shaft was also sunk deeper for drainage.

And now the actual boring of the tunnel commenced. It was to be 38 feet wide and 22½ feet in height. On New Year's Day, 1826, the boring-shield was placed below in the shaft. The shield was composed of 36 cells, 3 cells in height and 12 in breadth, with a workman to each.

The huge "shield" was placed before the earth to be excavated, and a front board being removed, the soil behind it was dug out to a specified extent, and the board was propped against the fresh surface thus made. When the boards had all been placed thus, the cells were pushed forward into the hollow then made. This was accomplished by means of screws at the top and bottom of the shield, and which were set against the completed brickwork behind.

For, while the labourers were working in front, the bricklayers behind built up the sides and roof, and formed the floor of the tunnel, the soil at the roof being supported by the shield until the masons had completed their task.

For nine feet, the tunnel proceeded through clay, but then came an unwelcome change. Wet, loose sand prevailed, and the work progressed with peril for thirty-two days, when firmer ground was reached. Six months passed and substantial headway was made, the tunnel being completed to the extent of 260 feet.

Then, on the 14th of September, the startling intelligence came that the engineer feared the river would burst in at the next tide. He had found a cavity over the shield. Sure enough, at high tide, when the river was brimming full, the workmen heard the ominous rattle of earth falling on their shield, while gushes of water followed.

So excellent were the precautions, however, that no disastrous effects followed, and Father Thames himself rolled earth or clay into the hole and stopped it up. It was a warning, and emphasised the fear that haunted the men's minds all through the hazardous undertaking—the fear that the river would break through and drown the tunnel.

In October, another small irruption took place, and was successfully combated. Then, in the following January (1827), some clay fell, but still no overwhelming catastrophe occurred. The ground grew so moist, however, that it was examined on the other side. That is, the river bed was inspected by the agency of a diving-bell, and some ominous depressions were found. These were promptly filled by bags of clay.

It may be asked, Why had Brunel not gone deeper? Why had he not placed a greater thickness of earth or clay between his work and the waters of the Thames?

The answer is this—He had been informed by geologists that quicksand prevailed lower down, and the shaft that he sank for drainage below the level of the proposed tunnel, indicated that this view might be correct. In fact, when he got down 80 feet, the soil gave way, and water and sand rushed upwards. He was therefore apparently between the Thames and the quicksand. The Tower Subway, constructed in 1869, and driven through the solid London clay, is, however, 60 feet deep where it commences at Tower Hill.

Work went steadily forward at Brunel's tunnel until the 18th of May. Mr. Beamish, the assistant engineer, was in the cutting on that day, and as the tide rose he observed the water increase about the shield; clay showed itself and gravel appeared. He had the clay closed up, and went to encourage the pumpers. Suddenly, before he could get into the cells, a great rush of sludge and water drove the men out of the cells, extinguished the lights, floated the cement casks and boxes, and poured forward and ever forward, filling the tunnel with the roaring of the flood.

The Thames had broken in with a vengeance this time, and drowned the tunnel.

CHAPTER II.

UNDER THE RIVER.

Happily no one lost his life.

The men retreated before the advancing wave, and as they went they met Brunel. But the great engineer could do nothing just then, except, like them, to retreat. The lights yet remaining flashed on the roaring water, and then suddenly went out in darkness.

The foot of the staircase was reached, and it was found thronged with the retreating workers. Higher and higher grew the surging flood; Brunel ordered great

speed; and scarcely were the men’s feet off the lower stair when it was torn away.

On gaining the top, cries were heard; some calling for a rope, others for a boat. Some one was below in the water! Brunel himself slipped down an iron rod, another followed, and each fastening a rope to the body of a man they found in the flood, he was soon drawn out of danger. On calling the roll, every worker answered to his name. No life was lost.

So far, good; but what was to be done now? The tunnel was full of water. To pump it dry was impossible, for the tide poured in from the Thames.

Again the diving-bell was used, and the hole was found in the bed of the river. To stop it bags of clay, with hazel sticks, were employed; and so difficult was the task that three thousand bags were utilised in the process, and more than a month elapsed before the water was subdued. Two months more passed before the earth washed in was removed, and Brunel could examine the work.

He found it for the most part quite sound, though near the shield it had been shorn of half its thickness of bricks. The chain of the shield was snapped in twain, and irons belonging to the same apparatus had been forced into the earth.

The men now proceeded with their task, and exhibited a cool courage deserving of all praise. Earth and water frequently fell; foul gases pervaded the stifling air, and sometimes exploded, or catching fire, they would now and again dance over the water; and again and again labourers would be carried away insensible from the poisonous atmosphere. Complaints, such as skin eruptions, sickness, and headaches, were common. Yet, in spite of every difficulty, the men worked on in that damp and dripping and fœtid mine, haunted ever with the dread of another flood.

And it came. On the 12th of August, 1828, some fifteen months after the previous disaster, the ground bulged out, a large quantity fell, and a violent rush of water followed; one man being washed out of his cell to the wooden staging behind.

[Illustration: THE THAMES TUNNEL.]

The flow was so great that Brunel ordered all to retire. The water rose so fast that when they had retreated a few feet it was up to their waists, and finally Brunel had to swim to the stairs, and the rush of water carried him up the shaft. Unhappily, about half-a-dozen lives were lost at this catastrophe, and those who were rescued—about a dozen in number—were extricated in an exhausted or fainting state. The roar of the water in the shaft made a deafening noise; the news soon spread, and the scene became very distressing as the relatives of the men arrived.

Once more the hole in the bed of the Thames had to be stopped. Down went the diving-bell, but it had to descend twice before the gap was discovered. It was a hole some seven feet long, and four thousand tons of earth, chiefly bags of clay, were used in filling it. Again the tunnel was entered, and again the intrepid engineer found the work sound.

But, alas, another difficulty had presented itself—one more difficult to conquer even than stopping up huge holes in the bed of the Thames. The tunnel was being cut

by a Company, and its money had gone; nay, more, its confidence had well nigh gone also. Work could not proceed without money, and for seven years silence and desolation reigned in those unfinished halls beneath the river.

Then the Government agreed to advance money, and work was again commenced. But it proceeded very slowly, some weeks less than a foot being cut, during others again three feet nine inches. The ground was in fact a fluid mud, and the bed of the river had to be artificially formed before the excavation could proceed in comparative safety. Further, the tunnel was far deeper than any other work in the neighbourhood, and all the water drained there—a difficulty which was obviated by the construction of a shaft on the other side of the river.

The shield had also to be replaced. It had been so battered about by the flood that another was necessary. As it kept up the earth above, and also in front, the change was both arduous and perilous. But it was accomplished without loss of life.

Three more irruptions of water occurred: the third in August, 1837, the fourth in November, 1837, and the fifth in March, 1838. But the engineer was more prepared for Father Thames' unpleasant visits, and a platform had been constructed by which the men could escape. Unhappily, one life was lost, however, on the fourth occasion. A great rush of soil also occurred in April, 1840, accompanied by a sinking of the shore at Wapping over some seven hundred feet of surface. Happily this occurred at low tide, and the chasm was filled with gravel and bags of clay before the river rose high.

At length, on the 13th of August, 1841, Brunel descended the shaft at Wapping, and entering a small cutting, passed through the shield in the tunnel, amidst the cheers of the workmen. After all these years of arduous toil, of anxious solicitude, and of hair-breadth escapes, the end was near, and a passage under the Thames was cut. It was not completed and open to the public, however, until the 25th of March, 1843, and then for foot passengers only.

The approaches for carriages remained to be constructed, and would have been expensive works. They were to be immense circular roads, but they were never made. Perhaps that deficiency contributed to the commercial failure of the great engineering enterprise. In any case, the tunnel never paid; the Company dissolved; and the tunnel passed over to the East London Railway, who run trains through it. Its length is 1300 feet, while between it and the river there is a thickness of soil of some fifteen feet.

Though a failure as a business, yet the tunnel was a great engineering triumph. It was a marvel of perseverance, and of determined, arduous, skilful toil against overwhelming difficulties. Eighteen years passed before it was completed; and if the seven be deducted during which the work was stopped, still eleven remain as the period of its construction. Work occupying such a length of time must be costly. Could it be shortened? Would tunnel-making machinery be developed and improved so as to expedite the labour of years?

CHAPTER III.

THROUGH THE ALPS.

“Cut through the Alps? It is an impossibility; and it would never pay!”

“Yet they are about to do it. Sommeiller, an engineer, has invented, or obtained, a rock-boring machine which promises to lighten the labour considerably; and then, of course, they will shatter great quantities of earth by explosives.”

“And what part of the Alps?”

“Through Mont Cenis. The tunnel will be about 7½ miles long, and the mountain over it will rise 5400 feet at one point.”

“And when do they expect to finish it?”

“I cannot say. They will begin on the southern—that is, the Italian—side first, and later on the French side. Through the tunnel will pass one of the principal routes from the West to the East.”

This conversation, we may suppose, took place in 1857, the year when the tunnel was commenced. For four years hand work was used, though blasting was in operation from the first; but in 1861 drilling by machinery was brought into play, and the rate of progress became much greater.

The machine was the first practical boring apparatus for rock, and was used first in making the Mont Cenis Tunnel. With explosives, as gun-cotton, dynamite, etc., the time occupied in cutting tunnels has been much reduced. Thus the Mont Cenis Tunnel occupied about thirteen years, and cost three millions of pounds. The St. Gotthard—another Alpine subway—occupied eight years, though it is 9¼ miles in length; and the Arlberg—yet another Alpine tunnel—a little over 6 miles long, occupied something more than three years.

Further, the railway of which the St. Gotthard Tunnel forms part, has been commercially very successful. This tunnel was commenced in 1872 and completed in 1880, the same year that saw the beginning of the Arlberg.

Tunnels through hard rock do not always need a lining of brickwork; but if the soil be clay, or loose earth of any kind, the lining of brick or stone must be brought up close to the scene of actual excavation. The Mont Cenis is lined with stone or brick almost entirely, about 900 feet, however, being without such lining.

And now, how was the actual work of tunnelling carried on? It will be seen at once that the problem was quite different from that of boring fifteen feet under the Thames, and sometimes through watery mud. In boring through mountains the quickest way of cutting and carting away rock is one of the chief points to be considered. At the Mont Cenis Tunnel the blasting took place by driving a series of shot holes into the soil, all over the surface to be cut, filling them with explosives, and firing them simultaneously in rings. Such explosives may be fired by a time-fuse or by electricity, giving the workmen ample time to escape out of reach. The shaken and shattered soil can then be cleared away.

The blast holes in this small-shot system are about 1 to 1½ inch in diameter, and from 1½ to 7 or 9 feet in the rock. The explosive is forced to the end of each, and the hole is then tamped—that is, closed with clay or sand—and fired in due time.

[Illustration: BORING MACHINE USED FOR THE MONT CENIS TUNNEL.]

The cutters for boring in rock are often diamond drills, the cutting edges being furnished with a kind of diamond found in Brazil, of a black colour and of great hardness. These are placed round the edge of a cylinder of steel, to which iron pipes can be screwed as the edge cuts its way deeper in the rock. The stuff cut out as the drill revolves finds its way through the cylinder and the piping. There are, however, a great number of boring machines of different kinds, hard steel sometimes taking the place of the opaque diamonds for cutting purposes. The compressed air with which many of the machines are worked assisted in the St. Gotthard in the ventilation of the tunnel, frequently a great consideration, as the space is so small and the gas from explosions often so great.

The Mont Cenis Tunnel marks a transition period in tunnelling. During the four years that hand labour was used, the average rate of progress was but nine inches a-day on either side; but when the rock-drills worked by compressed air were introduced, the speed was five times as great. Still further, at the Arlberg Tunnel through the Tyrolese Alps the average rate of progress was 9·07 yards per day, and the cost £108 per lineal yard; while the cost of the Mont Cenis was £226 per lineal yard. These figures show immense progress in economy and in speed.

The St. Gotthard Tunnel was begun in 1872, and the machine drills were used throughout. A heading was first cut about eight feet square, and the hollow thus gained was afterwards enlarged and finally sunk to the desired level. Several Ferroux drills were used, placed on a carriage, and an average charge of 1¾ lbs. of dynamite placed in the holes made. After firing, the compressed air was discharged and the shattered soil was cleared away.

In the Arlberg Tunnel a chief heading was driven, and then shafts opened up enabling smaller headings to be driven on both hands. Drills worked by hydraulic power were used, as well as drills worked by air, and, after the explosions, water spray was thrown out to assist in clearing and purifying the air. Ventilators also were used, which injected air at the rate of more than 8000 cubic feet per minute. Speedy transit of the earth excavated and the materials for masonry were also effected, it being estimated that some 900 tons of earth had to be taken out of each end, and about 350 tons of masonry had to be brought in, every day.

Tunnels through huge thicknesses of rock or under rivers can only be cut from the two opposite ends. Where possible, however, other shafts have been sunk along the line the subway was to take, and thus excavation might continue at several places along the line of route, the shafts being used for ventilation and for the conveyance of the excavated soil.

But the use of machine drills and of blasting explosives, with improved appliances for ventilation, have, with possibly some rare exceptions, rendered these methods obsolete. According to Pliny the tunnel for draining Lake Fucino was the greatest

work of his day. It was over 3½ miles long, and cut under Monte Salviano. Forty shafts were sunk in cutting it, also sloping galleries, and huge copper buckets were used to carry away the earth. It is stated that this tunnel—some ten feet high, by six wide—occupied 30,000 men eleven years. Compare this with the Arlberg, or even the Gotthard, double and treble the length, occupying much less time. Sir Benjamin Baker has calculated that the Fucino tunnel could now be cut in eleven months.

Gunpowder gave some advance on old Roman methods of tunnelling. The improved explosives and rock-drills have gone further.

Even as the Mont Cenis shows a transition period, so the Arlberg may be said to emphasise a triumph of the methods then indicated. So great have been the improvements of the rock-boring machinery, of the power of the blasts, and the speedy ventilation following the explosions, and of the quick transit of materials, that we shall most likely hear no more of sinking numerous shafts along the route.

But what of subaqueous tunnels? Violent explosives are hardly suitable for excavation a few feet under a turbid river. What is to be done, when cutting under a full and treacherous stream?

CHAPTER IV.

UNDER WATER AGAIN.

“How to cross the Thames at Blackwall, far east of the Tower Bridge?” That was a problem which the citizens of London had to face in the latter part of the nineteenth century.

An immense population dwelt on either side, and some means of easy communication became a pressing necessity. Should it be effected by means of a bridge, fixed or floating, or by means of a tunnel?

Finally a tunnel was decided upon, with sloping approaches on either side. Its entire length was to be 6200 feet including the approaches; but herein lay the danger and the difficulty—it was to be driven only seven feet below the bed of the river, and through loose soil and gravel.

How then was this perilous task to be accomplished? If the great river burst through Brunel’s fifteen feet, would it not be much more likely to rush through this seven feet of loose soil?

But the engineers in charge had an appliance in hand, which was unknown to Brunel—viz., a compressed air chamber, a piece of apparatus which has facilitated several great engineering achievements, besides the Blackwall Tunnel.

When the excavation of the tunnel was commenced, a stout apartment was formed at the end of the cutting, into which air was pumped until it exerted a pressure of some thirty-five pounds to a square inch, in addition to its usual weight.

This is generally reckoned at an average of 14·7 pounds to a square inch. We are so used to this pressure that we do not feel it; but let us enter a room where the air

has been much more compressed, as in this air-chamber, and serious consequences would be likely to ensue, especially at first.

The human body, however, has a wonderful power of adaptability, and after a time some men get used to the change and can work in the compressed air without injury. But at first it may cause bleeding from the nose and ears, sometimes indeed affecting the hearing more or less seriously, and also causing great pain.

The reason for using this compressed air chamber was to keep out Father Thames. The great pressure of the air resisted the great pressure of the water, and held up the seven feet of soil between.

Powerful engines were maintained at work to provide for the pressure of the air, and the chamber in which the compressed air was kept was entered and left by the workmen through an “air-lock”—that is, a small ante-chamber having two doors, one leading to the compressed air and the other to the ordinary atmosphere, and neither being opened at the same time.

The men, then, worked in this compressed air chamber, which prevented irruptions of the river. But the method of excavation was also another safeguard, both against irruptions of water and of earth.

In essence, it was much the same as that pursued in boring the tunnel for the South London Electric Railway; that, however, was through thick clay and about 10½ feet in diameter, and this was 27 feet across, and through loose and stony stuff. The shield, instead of containing as in Brunel’s time a number of cells, consisted of an immense iron cylinder, weighing some 250 tons; closed in front, but having a door in the closed part; the rim of the cylinder round this part having a sharp edge for cutting into the soil.

[Illustration: THE ENTRANCE TO THE AIR-LOCK.

(_Men waiting to enter the Compressed Air-Chamber through the Door._)]

The door being opened, the men found themselves face to face with the earth to be excavated. They cut away as well as they could, perhaps about 2½ feet deep, throwing the earth into trucks in the compressed air chamber; these trucks would be afterwards hauled away through the air-lock by electricity, and the huge iron cylinder would be pushed forward by means of hydraulic power. Twenty-eight hydraulic “jacks” were employed, and they forced forward the 250 ton cylinder with its cutting edge, when the men would resume working through the door as before.

Behind them, the hole of the tunnel thus cut out was being lined. First, it was built round with iron plates a couple of inches thick. This plating was fixed in segments, and formed a huge pipe a little smaller than the actual hollow in the earth. Through holes in the immense piping, liquid cement was forced, thus plugging up the space entirely between the earth and the iron, and forming an outer ring of cement.

Within, the tunnel was completed by a facing of glazed tiles, placed on a thickness of 14 inches of concrete. A road-way was laid 16 feet wide, flanked by footpaths of 3 feet, 2 inches, on either side. The subway is lighted by electricity, and staircases on

the banks lead down to it for foot passengers. The stairways give entrance to the tunnel not far from the river, and much nearer than the commencement of the carriage-way approaches.

At the northern side, the slope down commences near the East India Dock entrance, and turns out of the East India Dock Road. The slope is fairly gradual—about one in thirty-four—and it passes under the Blackwall line of the Great Eastern Railway, and near to Poplar Station. The part of the tunnel near to this point—that is the part between the river and the open slope—was executed by what is called “cut and cover” work—that is, a huge trench was dug, then arched in and covered over.

“Cut and cover” work also took place on the south side; and there, at the foot of an immense excavation ninety feet down, and with its sides held up by huge timbers, might have been seen a river of water which had drained in and was being pumped up quickly by powerful machinery.

Not far distant, the shaft was being sunk for the staircase. In principle, the sinking of the shaft was conducted much as Brunel’s shaft at the Thames Tunnel, only it was built up of iron instead of brick. Imagine a big gasometer with a scaffold near the top, where men are busy building the walls higher and higher by adding on plate after plate of iron. On reaching the scaffold you find that there are two great cylinders of iron, one standing inside the other, and concrete is being filled in between them. Men also are down below digging out the earth which is being swung up in iron buckets; and as the soil is gradually removed, the immense double iron and concrete cylinder slowly sinks by its own weight.

In this manner, the great shaft was sunk nearly ninety feet, and within it the staircase has been built, giving entrance for foot passengers, not far from the river. Thus, on either side are sloping entrances to the tunnel, and also, nearer the water, stairways of descent down great shafts.

Engineers have also found their way beneath other great English rivers—the Severn and the Mersey. Much water had to be dealt with in the cutting of the Severn Tunnel. This important work, four and one-third miles long, was driven in some places forty-five feet under sandstone, and at the Salmon Pool—a hollow in the river bed—the tunnel was thirty feet under soil called trias marl. Much greater space, therefore, exists here between the tunnel and river than at Blackwall. But the river burst through. The work was begun in 1873, and completed in 1886.

Six years after its commencement the tunnel was drowned, so to speak, for a long time by a large spring of water which burst out from limestone, and arrangements had to be made to provide for this flood. It is now conducted by a subsidiary tunnel or channel to a huge shaft, where it is raised by pumps of sufficient strength. Then there was the perilous Salmon Pool to be dealt with. The river burst through here, and the rent had to be stopped with clay. The tunnel is twenty-six feet wide by twenty feet high, and is cut through Pennant stone, shale, and marl. It is lined with Staffordshire vitrified bricks throughout—seventy-five million bricks it is estimated being used. The works are ventilated by a huge fan, and pumping continually proceeds, something

like twenty-six million gallons of water, it is said, being raised in the twenty-four hours. The tunnel, of which the engineers were Messrs. Hawkshaw, Son, Hayter & Richardson, and Mr. T. A. Walker, Contractor, is for the use of the Great Western Railway, and saves that Company's Welsh and Irish trains to Milford a long way round by Gloucester.

[Illustration: THE BORING MACHINE USED IN THE PRELIMINARY CONSTRUCTION OF THE ENGLISH CHANNEL TUNNEL.]

In cutting the Mersey Tunnel, which was completed in 1886, machinery was used for some of the work. The machine bored partly to a diameter of seven feet four inches, but hand labour had to be largely depended upon. The plan pursued was to sink a shaft on either side of the river and drive a heading, sloping upward through the sandstone to the centre; this heading acting as a drain for any water which might appear. The thickness between the arch of the tunnel and the river bed is thirty feet at its least, and the tunnel, which occupied about six years in construction, and of which the engineers were Messrs. Brunlees & Fox, is provided with pumps raising some thirteen million gallons of water daily. As in the case of the Severn Tunnel, ventilation is provided for by huge fans.

A boring machine was also used in the preliminary efforts for the construction of a tunnel under the English Channel. Holes, seven feet across and to the length of 2000 yards, have been bored by a compressed air machine, working with two arms furnished with teeth of steel. The construction of the tunnel is held to be quite feasible from an engineering point of view, and it is believed that it would pass through strata impervious to water, such as chalk marl and grey chalk.

Still, the huge tunnel at Blackwall, which was carried out by Mr. Binnie, Chief Engineer of the London County Council, with Mr. Greathead and Sir Benjamin Baker as Consulting Engineers, is probably one of the most daring and stupendous enterprises of the kind ever undertaken. To hollow out a subway hundreds of feet long under the Thames, only seven feet from the bed of the great river, and through loose gravelly soil, was a great triumph. It was achieved not by uncalculating bravery, but by a wise combination of cool courage, superb skill, and admirable foresight.

To design effectively, to provide for contingencies, to be daunted by no difficulties—these qualities help to produce the Triumphs of Engineers, as well as do great inventive skill, the power of adapting principles to varying circumstances, and high-spirited enterprise in planning and conducting noble and useful works. These works may well rank among the great achievements of man's effort and the wonders of the world.

THE END.

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